

Topic 2

Complex Geometry and the Kähler Hodge Decomposition

Hodge Theory

A complex structure gives types. The Kähler condition lets cohomology inherit them.

The question. Lecture 1 identifies de Rham cohomology with harmonic forms after choosing a metric. On a complex manifold, differential forms have an additional decomposition by type. When does this pointwise decomposition descend to cohomology, and what connects this question with a connection and its curvature?

Route through the lecture. We construct the complex and Dolbeault complexes, introduce Hermitian geometry, and derive the Chern connection from its defining properties. The Kähler condition removes the tangent connection's torsion and makes the local operator identities work. These identities identify the de Rham and Dolbeault Laplacians up to a factor of two. The analytic theorem from Lecture 1 then yields the Hodge decomposition of complex cohomology. Complex tori provide explicit coordinates, connections, Fourier solutions, harmonic bases, and period computations throughout. Curves, the sphere as $\mathbb{P}^1$, projective space, hyperbolic quotients, and a Hopf surface make the hypotheses visible.

Conventions. The complex dimension is n and the real dimension is $2n$. Unless specified otherwise, forms and cohomology coefficients are complex. Hermitian inner products are linear in the first variable. The Hodge star is extended complex linearly, so

$$(\alpha, \beta)_{L^2} = \int_X \alpha \wedge \bar{*}\beta, \quad \omega_0 = \frac{i}{2} \sum_j dz_j \wedge d\bar{z}_j, \quad \Delta_d = dd^* + d^*d \geq 0.$$

Our square elliptic curve is $E = \mathbb{C}/(\mathbb{Z} + i\mathbb{Z})$ with coordinates of period one. Global Hodge theorems assume compactness and no boundary. These hypotheses are repeated at each theorem; local geometry and local identities do not require compactness.

How to read. This is an expanded reference for the second lecture. The blackboard companion selects a classroom route and gives solutions to all twenty-two exercises. Elementary deductions and the Chern and Kähler calculations are proved. The local $\bar{\partial}$ existence input, integrability theorem, and elliptic regularity input are identified when used. Lecture 1 supplies the underlying smooth Hodge theory.

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1 Complex manifolds and the Dolbeault complex

Lecture 1 used the differential d to turn smooth forms into topological invariants. A complex structure divides a form into types and divides d into two operators. Before relating the resulting cohomology groups, we must separate three inputs: holomorphic coordinates, a Hermitian metric, and the extra differential condition that makes the metric Kähler. We write $\mathcal{A}^\bullet(X; \mathbb{C})$ for the smooth de Rham complex and $\mathcal{A}^{p,q}(X)$ for its smooth forms of type (p, q) . The notation Ω_X^p is reserved for the sheaf of holomorphic forms. In a later comparison for a smooth complex algebraic variety Y , we will distinguish the algebraic sheaf complex $\Omega_{Y/\mathbb{C}}^\bullet$ from the holomorphic sheaf complex $\Omega_{Y^{\text{an}}}^\bullet$ and the global smooth complex $\mathcal{A}^\bullet(Y^{\text{an}}; \mathbb{C})$. Algebraic de Rham cohomology is the hypercohomology of its sheaf complex; global algebraic forms alone need not suffice.

All manifolds below are smooth, Hausdorff, and second countable. A compact manifold has no boundary unless a boundary is explicitly mentioned.

1.1 Holomorphic coordinates and the running torus example

Definition 1.1. A *complex manifold of complex dimension n* is a real $2n$ -manifold with charts $z = (z^1, \dots, z^n)$ taking values in $\mathbb{C}^n$, whose transition maps are holomorphic. A function is holomorphic if its expression in every such chart is holomorphic. A map of complex manifolds is holomorphic if its coordinate expressions are holomorphic.

Write $z^j = x^j + iy^j$. Multiplication by i on coordinate tangent spaces defines a real bundle map

$$J\partial_{x^j} = \partial_{y^j}, \quad J\partial_{y^j} = -\partial_{x^j}, \quad J^2 = -\text{id}.$$

The Cauchy–Riemann equations say that the derivative of a holomorphic transition map commutes with J . Thus these locally defined maps patch. The ordered real coordinates $(x^1, y^1, \dots, x^n, y^n)$ give the *complex orientation*; holomorphic coordinate changes preserve it because their real Jacobian determinant is the squared absolute value of their complex determinant.

Remark 1.2 (An almost complex structure is less data than a complex atlas). A smooth map $J : TX \rightarrow TX$ with $J^2 = -\text{id}$ is called an *almost complex structure*. Such a J need not come from holomorphic coordinates. The Newlander–Nirenberg theorem says that it does precisely when its Nijenhuis tensor vanishes, equivalently when $T^{0,1}X$ is closed under Lie brackets. We use this theorem only to name the integrability issue; our manifolds come equipped with a complex atlas. In particular, the identity $d = \partial + \bar{\partial}$ proved below is being asserted on complex manifolds, not on arbitrary almost complex manifolds.

Example 1.3 (A complex torus). Let $\Lambda \subset \mathbb{C}^n$ be a lattice of real rank $2n$. Small open balls in $\mathbb{C}^n$ project injectively to $X = \mathbb{C}^n/\Lambda$, and two inverse charts differ by $z \mapsto z + \lambda$. Since translations are holomorphic, X is a compact complex manifold. The functions z^j are not global functions on X , whereas dz^j and ∂_{z^j} are global: their expressions are unchanged by a lattice translation.

For $E = \mathbb{C}/(\mathbb{Z} + i\mathbb{Z})$, a smooth function is a smooth function of (x, y) that is periodic with period 1 in both variables. The form $dz = dx + i dy$ has periods 1 and i around the horizontal and vertical circles. It is a global holomorphic one-form, but it cannot be the differential of a global function. Thus the torus already separates a globally defined differential from a globally defined coordinate.

Example 1.4 (The Riemann sphere). The projective line $\mathbb{P}^1$ has two charts $z = Z_1/Z_0$ and $w = Z_0/Z_1$, with $w = 1/z$ on their overlap. The transition is holomorphic, and

$$dz = -w^{-2} dw.$$

Consequently dz on the affine chart does not extend to a holomorphic one-form at infinity. More generally, if $f(z) dz$ extends holomorphically to all of $\mathbb{P}^1$, then f is entire and $-w^{-2}f(1/w)$ is holomorphic near $w = 0$. Hence $f(z) = O(|z|^{-2})$ at infinity; Liouville's theorem forces $f = 0$. The sphere has no nonzero holomorphic one-form, in contrast with E .

1.2 Type decomposition: linear algebra at each point

Extend J complex-linearly to $T_{\mathbb{C}}X = TX \otimes_{\mathbb{R}} \mathbb{C}$. Its eigenspaces give a direct sum

$$T_{\mathbb{C}}X = T^{1,0}X \oplus T^{0,1}X, \quad T^{1,0}X = \ker(J - i), \quad T^{0,1}X = \ker(J + i).$$

In a holomorphic chart,

$$\partial_{z^j} = \frac{1}{2}(\partial_{x^j} - i\partial_{y^j}), \quad \partial_{\bar{z}^j} = \frac{1}{2}(\partial_{x^j} + i\partial_{y^j})$$

span these two bundles. Their dual frames are $dz^j = dx^j + i dy^j$ and $d\bar{z}^j = dx^j - i dy^j$. Notice both the factor $1/2$ for vectors and its absence for covectors: $dz^j(\partial_{z^k}) = \delta_{jk}$.

Let $\mathcal{A}^{p,q}(X)$ denote the space of smooth forms that locally are sums

$$\alpha = \sum_{|I|=p, |J|=q} \alpha_{I\bar{J}} dz^I \wedge d\bar{z}^J,$$

where the indices in I and J are increasing. Then

$$\mathcal{A}^k(X; \mathbb{C}) = \bigoplus_{p+q=k} \mathcal{A}^{p,q}(X), \quad \overline{\mathcal{A}^{p,q}(X)} = \mathcal{A}^{q,p}(X). \quad (1.1)$$

Wedge product adds bidegrees. This is a decomposition of bundles and of smooth forms; no metric and no cohomology theorem have been used.

Example 1.5 (Real forms and pure types). On $\mathbb{C}$, $dz \wedge d\bar{z} = -2i dx \wedge dy$, so $dx \wedge dy = (i/2) dz \wedge d\bar{z}$ has type $(1, 1)$. On $\mathbb{C}^2$, direct substitution gives

$$\begin{aligned} dx^1 \wedge dx^2 + dy^1 \wedge dy^2 &= \frac{1}{2}(dz^1 \wedge d\bar{z}^2 - dz^2 \wedge d\bar{z}^1), \\ dx^1 \wedge dx^2 - dy^1 \wedge dy^2 &= \frac{1}{2}(dz^1 \wedge dz^2 + d\bar{z}^1 \wedge d\bar{z}^2). \end{aligned}$$

The first real form is of type $(1, 1)$; the second has types $(2, 0)$ and $(0, 2)$. A nonzero real form of pure type must have $p = q$, because complex conjugation fixes a real form and exchanges the two indices. All these constant forms descend to the standard complex two-torus.

1.3 The two differentials and their signs

For a smooth function,

$$df = \sum_j f_{z^j} dz^j + \sum_j f_{\bar{z}^j} d\bar{z}^j, \quad f_{z^j} = \frac{1}{2}(f_{x^j} - i f_{y^j}), \quad f_{\bar{z}^j} = \frac{1}{2}(f_{x^j} + i f_{y^j}).$$

Since the coordinate one-forms are closed, differentiating a (p, q) -form produces only types $(p+1, q)$ and $(p, q+1)$. The corresponding projections define ∂ and $\bar{\partial}$:

$$\begin{aligned} d &= \partial + \bar{\partial}, \\ \partial\alpha &= \sum_{I, J, j} (\alpha_{I\bar{J}})_{z^j} dz^j \wedge dz^I \wedge d\bar{z}^J, \\ \bar{\partial}\alpha &= \sum_{I, J, j} (\alpha_{I\bar{J}})_{\bar{z}^j} d\bar{z}^j \wedge dz^I \wedge d\bar{z}^J. \end{aligned} \quad (1.2)$$

The last expression includes the signs arising when $d\bar{z}^j$ is moved past dz^I . Both operators satisfy the graded Leibniz rule with the *total* degree of the first factor.

Proposition 1.6. *On a complex manifold,*

$$\partial^2 = 0, \quad \bar{\partial}^2 = 0, \quad \partial\bar{\partial} + \bar{\partial}\partial = 0.$$

Proof. Expand $0 = d^2 = \partial^2 + (\partial\bar{\partial} + \bar{\partial}\partial) + \bar{\partial}^2$. The three terms have distinct type increments $(2, 0), (1, 1), (0, 2)$. The type decomposition is a direct sum, so each term vanishes. $\square$

Example 1.7 (A scalar calculation and a sign check). For $f = z^2\bar{z} + e^z$ on $\mathbb{C}$,

$$\partial f = (2z\bar{z} + e^z) dz, \quad \bar{\partial} f = z^2 d\bar{z}, \quad \partial\bar{\partial} f = 2z dz \wedge d\bar{z} = -\bar{\partial}\partial f.$$

For $\alpha = \bar{z}^2 dz^1$ on $\mathbb{C}^2$, where $\bar{z}^2$ denotes the second coordinate rather than a square, $\bar{\partial}\alpha = d\bar{z}^2 \wedge dz^1 = -dz^1 \wedge d\bar{z}^2$. Holomorphicity of a $(p, 0)$ -form means that its coefficient functions are holomorphic, equivalently $\bar{\partial}\alpha = 0$. It need not imply $d\alpha = 0$: the holomorphic one-form $z^1 dz^2$ on $\mathbb{C}^2$ satisfies $d(z^1 dz^2) = dz^1 \wedge dz^2$. Compactness will later change this conclusion in the Kähler case.

Definition 1.8 (Dolbeault cohomology). For each fixed p , the complex

$$0 \longrightarrow \mathcal{A}^{p,0}(X) \xrightarrow{\bar{\partial}} \mathcal{A}^{p,1}(X) \xrightarrow{\bar{\partial}} \cdots \xrightarrow{\bar{\partial}} \mathcal{A}^{p,n}(X) \longrightarrow 0$$

has cohomology

$$H_{\bar{\partial}}^{p,q}(X) = \frac{\ker(\bar{\partial} : \mathcal{A}^{p,q} \rightarrow \mathcal{A}^{p,q+1})}{\text{im}(\bar{\partial} : \mathcal{A}^{p,q-1} \rightarrow \mathcal{A}^{p,q})}.$$

Spaces with an index outside $0, \dots, n$ are understood to be zero. In particular $H_{\bar{\partial}}^{p,0}(X)$ is the vector space of global holomorphic p -forms; there is no quotient in antiholomorphic degree zero.

1.4 Local solvability, a fine resolution, and a global obstruction

Theorem 1.9 (Local Dolbeault lemma). *If $q > 0$ and $\bar{\partial}\alpha = 0$ for a smooth (p, q) -form, then near every point there is a smooth $(p, q-1)$ -form β with $\alpha = \bar{\partial}\beta$.*

Proof. The one-variable analytic input is the fundamental-solution identity

$$\frac{\partial}{\partial\bar{z}} \frac{1}{\pi z} = \delta_0 \quad \text{in distributions.}$$

Let $D' \Subset D$ be disks, and choose a smooth function χ compactly supported in D with $\chi = 1$ near $\overline{D'}$. For smooth f on D , put

$$Tf(z) = \frac{1}{\pi} \int_{\mathbb{C}} \frac{\chi(\zeta)f(\zeta)}{z - \zeta} dA(\zeta). \quad (1.3)$$

The kernel is locally integrable. Convolution with the compactly supported smooth function χf is smooth, and the displayed distribution identity gives $(Tf)_{\bar{z}} = \chi f = f$ on D' . The cutoff makes the extension outside D smooth and removes any unjustified boundary regularity assumption.

For $(0, q)$ -forms on a polydisk, argue by induction on its dimension. Write

$$\alpha = \alpha_0 + d\bar{z}^n \wedge \alpha_1,$$

where neither α_0 nor α_1 contains $d\bar{z}^n$. Apply the operator T in the variable z^n , coefficient by coefficient, and set $\beta_1 = T_n \alpha_1$. On a smaller polydisk, the form $\alpha - \bar{\partial}\beta_1$ has no $d\bar{z}^n$ component. It remains $\bar{\partial}$ -closed, so its coefficients are holomorphic in z^n and it is closed for the $\bar{\partial}$ operator in the first $n-1$ variables. The induction hypothesis solves it there. The same integral operators

in those variables preserve holomorphic dependence on z^n , since the cutoffs depend only on their respective integration variables. This produces β_0 with $\alpha - \bar{\partial}\beta_1 = \bar{\partial}\beta_0$ and proves the induction. If its antiholomorphic degree exceeds $n - 1$, that residual form is zero.

For general p , write $\alpha = \sum_I \alpha_I \wedge dz^I$ with α_I of type $(0, q)$. Since dz^I is closed, the same construction for each α_I gives the asserted primitive, with no further differentiation of the holomorphic coordinate frame. $\square$

Let Ω_X^p be the sheaf of holomorphic p -forms; in degree zero it is $\mathcal{O}_X$. Let $\mathcal{A}_X^{p,q}$ be the sheaf of smooth (p, q) -forms, so $\Gamma(X, \mathcal{A}_X^{p,q}) = \mathcal{A}^{p,q}(X)$. The local lemma, including the description of the degree-zero kernel, says that

$$0 \longrightarrow \Omega_X^p \longrightarrow \mathcal{A}_X^{p,0} \xrightarrow{\bar{\partial}} \mathcal{A}_X^{p,1} \xrightarrow{\bar{\partial}} \cdots \xrightarrow{\bar{\partial}} \mathcal{A}_X^{p,n} \longrightarrow 0 \quad (1.4)$$

is an exact sequence of sheaves. The smooth-form sheaves are fine because one can multiply their sections by smooth partitions of unity. Applying the acyclic resolution theorem from Lecture 1 over $\mathbb{C}$ gives the *Dolbeault theorem*

$$H^q(X, \Omega_X^p) \cong H_{\bar{\partial}}^{p,q}(X). \quad (1.5)$$

This comparison needs neither compactness nor a metric. The resolution starts with holomorphic forms, rather than the constant sheaf used in de Rham theory. It therefore computes different groups; identifying their relationship is the main theorem of this lecture.

Example 1.10 (Why a local Dolbeault primitive does not descend). On $E = \mathbb{C}/(\mathbb{Z} + i\mathbb{Z})$, $d\bar{z} = \bar{\partial}\bar{z}$ upstairs. The primitive $\bar{z}$ changes by 1 and by $-i$ under the two generating translations. Suppose a periodic function u satisfied $\bar{\partial}u = d\bar{z}$. Then $u_{\bar{z}} = \frac{1}{2}(u_x + iu_y) = 1$, but periodicity implies

$$\int_0^1 \int_0^1 u_{\bar{z}} dx dy = 0,$$

whereas the integral of 1 is 1. Thus $[d\bar{z}] \neq 0$ in $H_{\bar{\partial}}^{0,1}(E)$. Smooth partitions of unity can patch functions, but differentiating the patched function introduces derivatives of the partition. Those additional terms retain the obstruction, just as for the de Rham class of dx in Lecture 1.

1.5 Hermitian metrics, complex forms, and the Hodge star

A Riemannian metric g is *Hermitian* for J if $g(Ju, Jv) = g(u, v)$. Such metrics always exist: replace any Riemannian metric g_1 by $\frac{1}{2}(g_1(u, v) + g_1(Ju, Jv))$. Its *fundamental form* is

$$\omega(u, v) = g(Ju, v).$$

It is alternating because $g(Ju, v) = -g(u, Jv)$. In complex coordinates, we use coefficients $G_{i\bar{j}}$ normalized by

$$g(u, v) = \operatorname{Re} \sum_{i,j} G_{i\bar{j}} dz^i(u) \overline{dz^j(v)} \quad (u, v \text{ real}), \quad (1.6)$$

$$\omega = \frac{i}{2} \sum_{i,j} G_{i\bar{j}} dz^i \wedge d\bar{z}^j.$$

Here G is a positive Hermitian coefficient array. In particular $g_0 = \sum_j ((dx^j)^2 + (dy^j)^2)$ corresponds to $G_{i\bar{j}} = \delta_{ij}$. The form ω is real, of type $(1, 1)$, and positive in the sense $\omega(u, Ju) = g(u, u) > 0$ for nonzero real u . At each point,

$$\operatorname{vol}_g = \frac{\omega^n}{n!}. \quad (1.7)$$

This identity is pointwise linear algebra and holds for every Hermitian metric, whether or not $d\omega = 0$.

Extend the real inner product on forms to a Hermitian inner product *linear in the first variable*, conjugate-linear in the second. For decomposable forms its definition is the determinant

$$\langle \alpha_1 \wedge \cdots \wedge \alpha_k, \beta_1 \wedge \cdots \wedge \beta_k \rangle_g = \det(\langle \alpha_a, \beta_b \rangle_g)_{a,b=1}^k.$$

For the flat metric, $|dz^j|^2 = |d\bar{z}^j|^2 = 2$ and $\langle dz^j, d\bar{z}^k \rangle = 0$. Thus $\theta^j = dz^j/\sqrt{2}$ and $\bar{\theta}^j = d\bar{z}^j/\sqrt{2}$ form a unitary coframe of the complexified cotangent space. More generally a unitary coframe can be chosen at every point, in which

$$\omega = i \sum_j \theta^j \wedge \bar{\theta}^j.$$

The forms $\theta^I \wedge \bar{\theta}^J$ form an orthonormal basis. Consequently different bidegrees are orthogonal.

The real Hodge star from Lecture 1 is extended *complex-linearly*. Then

$$\alpha \wedge *\bar{\beta} = \langle \alpha, \beta \rangle_g \text{vol}_g, \quad *^2|_{\mathcal{A}^k} = (-1)^{k(2n-k)}, \quad *\bar{\alpha} = \overline{*\alpha}. \quad (1.8)$$

Complementing the indices in a unitary coframe gives

$$* : \mathcal{A}^{p,q} \longrightarrow \mathcal{A}^{n-q,n-p}. \quad (1.9)$$

One can also see the indices from (1.8): to pair a (p, q) -form with $*\bar{\beta}$, the latter must have type $(n-p, n-q)$. Thus the complex-linear operator $*$ and the conjugate-linear operator $\alpha \mapsto *\bar{\alpha}$ have different type rules.

On a compact X , the global inner product is

$$(\alpha, \beta)_{L^2} = \int_X \langle \alpha, \beta \rangle_g \text{vol}_g = \int_X \alpha \wedge *\bar{\beta}.$$

Its completion gives the Hilbert space $L^2\mathcal{A}^k(X; \mathbb{C})$. On a noncompact manifold the same definition applies to square-integrable forms, but an arbitrary smooth form need not be in this space.

Example 1.11 (One complex dimension, including a conformal torus). Let $g = \rho^2(dx^2 + dy^2)$ on a complex coordinate chart, where $\rho > 0$. Then

$$\begin{aligned} \omega &= \frac{i\rho^2}{2} dz \wedge d\bar{z}, & *dz &= -i dz, & *d\bar{z} &= i d\bar{z}, \\ *1 &= \omega, & *(dz \wedge d\bar{z}) &= -2i\rho^{-2}, & |dz|_g^2 &= 2\rho^{-2}. \end{aligned}$$

The star on one-forms is unchanged by the conformal factor, although the pointwise norm and area form each change. Their changes cancel in the integral:

$$\|f dz\|_{L^2}^2 = 2 \int |f|^2 dx dy.$$

When ρ is periodic, these formulas hold on the elliptic curve E . They explain why conformal changes will leave its harmonic one-forms fixed while changing its harmonic volume form.

1.6 The Dolbeault Laplacian and the analytic input

On compact X , define the formal adjoint $\bar{\partial}^*$ by $(\bar{\partial}\alpha, \beta)_{L^2} = (\alpha, \bar{\partial}^*\beta)_{L^2}$. It maps (p, q) -forms to $(p, q-1)$ -forms. Stokes' theorem and the fact that the real dimension $2n$ is even give

$$\bar{\partial}^* = - * \partial *, \quad \partial^* = - * \bar{\partial} *.$$

These are local differential-operator identities and do not require the Kähler condition. Their interpretation as adjoint identities on a noncompact manifold requires compact support or appropriate domains. Define

$$\Delta_{\bar{\partial}} = \bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial}, \quad \mathcal{H}_{\bar{\partial}}^{p,q}(X) = \ker \Delta_{\bar{\partial}}|_{\mathcal{A}^{p,q}(X)}. \quad (1.10)$$

Integration by parts gives the energy identity

$$(\Delta_{\bar{\partial}}\alpha, \alpha)_{L^2} = \|\bar{\partial}\alpha\|_{L^2}^2 + \|\bar{\partial}^*\alpha\|_{L^2}^2. \quad (1.11)$$

It follows that a smooth form on compact X is Dolbeault harmonic exactly when it is both $\bar{\partial}$ -closed and $\bar{\partial}$ -coclosed.

Example 1.12 (Checking the factor of two on $\mathbb{C}$). For $g_0 = dx^2 + dy^2$, the star formulas give

$$\bar{\partial}^*(b d\bar{z}) = -2b_z, \quad \Delta_{\bar{\partial}}f = -2f_{z\bar{z}} = -\frac{1}{2}(f_{xx} + f_{yy}).$$

For example, $-\ast\partial\ast(b d\bar{z}) = -\ast(ib_z dz \wedge d\bar{z}) = -2b_z$. The real Hodge Laplacian on functions is $\Delta_d f = -(f_{xx} + f_{yy})$. The factor 2 in $\Delta_d = 2\Delta_{\bar{\partial}}$ therefore already appears in one flat chart; it comes from the normalization of dz , not from a change of sign convention. Later we will prove this identity on all forms of a Kähler manifold.

Here is why the elliptic analysis from Lecture 1 applies. For a real covector ξ , the first-order symbol of $\bar{\partial}$ is, up to the harmless factor i , wedge multiplication by $\xi^{0,1}$. Exterior multiplication and its adjoint satisfy

$$(\xi^{0,1} \wedge)(\xi^{0,1} \wedge)^* + (\xi^{0,1} \wedge)^*(\xi^{0,1} \wedge) = |\xi^{0,1}|_g^2 \text{id}.$$

Since ξ is real, its two type components have equal norm and are orthogonal. Thus

$$\sigma_2(\Delta_{\bar{\partial}})(x, \xi) = \frac{1}{2}|\xi|_g^2 \text{id}. \quad (1.12)$$

This is invertible for $\xi \neq 0$, so $\Delta_{\bar{\partial}}$ is elliptic. Only the principal symbol has been compared with that of Δ_d . On a general Hermitian manifold their lower-order terms need not agree.

Theorem 1.13 (The analytic package, recalled from Lecture 1). *For compact Hermitian X , the nonnegative self-adjoint realization of $\Delta_{\bar{\partial}}$ on L^2 has domain the Sobolev space H^2 and discrete spectrum, with finite-dimensional eigenspaces of smooth forms. Let $P_{\bar{\partial}}$ be orthogonal projection onto its kernel. There is a Green operator $G_{\bar{\partial}}$, zero on the kernel, such that*

$$\Delta_{\bar{\partial}}G_{\bar{\partial}} = G_{\bar{\partial}}\Delta_{\bar{\partial}} = \text{id} - P_{\bar{\partial}}.$$

It maps H^s continuously to H^{s+2} and maps smooth forms to smooth forms. In particular a smooth equation $\Delta_{\bar{\partial}}u = f$ is solvable precisely when f is orthogonal to all Dolbeault harmonic forms.

The black box is the elliptic estimate, regularity, and compactness needed to establish this package; see the analytic references in Lecture 1 and [1]. The reasoning behind finite-dimensionality is worth repeating. The elliptic estimate

$$\|u\|_{H^2} \leq C(\|\Delta_{\bar{\partial}}u\|_{L^2} + \|u\|_{L^2})$$

bounds the H^2 norm of the unit ball in the kernel. Compactness of $H^2 \hookrightarrow L^2$ then makes that unit ball compact in L^2 . An infinite orthonormal sequence would have pairwise distance $\sqrt{2}$

and no convergent subsequence, a contradiction. To invert away from the kernel, one similarly obtains a positive spectral gap, or equivalently

$$\|u\|_{L^2} \leq C \|\Delta_{\bar{\partial}} u\|_{L^2} \quad \text{when } u \perp \ker \Delta_{\bar{\partial}}.$$

If this estimate failed, normalized counterexamples, the elliptic estimate, and compactness would produce a nonzero harmonic limit orthogonal to the kernel. Self-adjointness and closed range then give the inverse on this orthogonal complement; elliptic regularity makes the solution smooth whenever the right-hand side is smooth.

Theorem 1.14 (Dolbeault Hodge theorem for a Hermitian metric). *On a compact Hermitian manifold,*

$$\mathcal{A}^{p,q}(X) = \mathcal{H}_{\bar{\partial}}^{p,q}(X) \oplus^{\perp} \bar{\partial} \mathcal{A}^{p,q-1}(X) \oplus^{\perp} \bar{\partial}^* \mathcal{A}^{p,q+1}(X). \quad (1.13)$$

Every Dolbeault class has a unique Dolbeault harmonic representative, so $\mathcal{H}_{\bar{\partial}}^{p,q}(X) \cong H_{\bar{\partial}}^{p,q}(X)$. In particular these cohomology groups have finite dimension. Their dimensions are denoted $h^{p,q}(X) = \dim_{\mathbb{C}} H_{\bar{\partial}}^{p,q}(X)$ and are called the Hodge numbers.

Proof. Apply the Green identity to a smooth form:

$$\alpha = P_{\bar{\partial}} \alpha + \bar{\partial} \bar{\partial}^* G_{\bar{\partial}} \alpha + \bar{\partial}^* \bar{\partial} G_{\bar{\partial}} \alpha.$$

Harmonic forms are orthogonal to the last two images by the energy identity. Moreover $(\bar{\partial} u, \bar{\partial}^* v)_{L^2} = (\bar{\partial}^2 u, v)_{L^2} = 0$. This proves the orthogonal direct sum. If $\bar{\partial} \alpha = 0$, then α is orthogonal to $\text{im } \bar{\partial}^*$, so its last component vanishes and $\alpha - P_{\bar{\partial}} \alpha$ is $\bar{\partial}$ -exact. If a harmonic form is also $\bar{\partial}$ -exact, it belongs to two orthogonal summands and is zero. These statements prove existence and uniqueness of the representative. $\square$

This theorem requires a Hermitian metric, but no Kähler assumption. It concerns $\bar{\partial}$ -cohomology. At this stage there is no assertion that a Dolbeault harmonic form is d -harmonic, or even d -closed. For $(p, 0)$ -forms, $\bar{\partial}^*$ is identically zero for degree reasons, so

$$\mathcal{H}_{\bar{\partial}}^{p,0}(X) = H^0(X, \Omega_X^p).$$

Every holomorphic function on compact connected X is constant: its absolute value attains a maximum, and the maximum principle in a coordinate ball, followed by connectedness, forces constancy. Hence $h^{0,0} = 1$. Compactness cannot be omitted from the energy argument. For example $\Delta_{\bar{\partial}} \bar{z} = 0$ on flat $\mathbb{C}$ although $\bar{\partial} \bar{z} = d\bar{z} \neq 0$; the global L^2 integration used above is unavailable for this function.

1.7 Fourier series: solving the Dolbeault problem on a torus

Example 1.15 (The square elliptic curve, with its flat metric). On $E = \mathbb{C}/(\mathbb{Z} + i\mathbb{Z})$, put $e_{mn}(x, y) = e^{2\pi i(mx + ny)}$. Direct differentiation gives

$$\begin{aligned} \partial e_{mn} &= \pi(im + n)e_{mn} dz, \\ \bar{\partial} e_{mn} &= \pi(im - n)e_{mn} d\bar{z}, \\ \Delta_{\bar{\partial}} e_{mn} &= 2\pi^2(m^2 + n^2)e_{mn}. \end{aligned}$$

The metric and frames are constant, so the same scalar eigenvalue applies to each coefficient of $e_{mn} dz^I \wedge d\bar{z}^J$. Only the zero Fourier mode is harmonic. Therefore

$$\begin{array}{c|cccc} (p, q) & (0, 0) & (1, 0) & (0, 1) & (1, 1) \\ \hline \mathcal{H}_{\bar{\partial}}^{p,q}(E) & \mathbb{C} \cdot 1 & \mathbb{C} \cdot dz & \mathbb{C} \cdot d\bar{z} & \mathbb{C} \cdot (dz \wedge d\bar{z}) \end{array} \quad (1.14)$$

For this metric, taking a harmonic representative means averaging each coefficient over the unit square. This is an explicit realization of the abstract projection $P_{\bar{\partial}}$.

We can solve the first nontrivial equation without quoting the abstract theorem. Write $f = \sum f_{mn} e_{mn}$. The equation $\bar{\partial}u = f d\bar{z}$ is solvable if and only if $f_{00} = 0$, and then its unique solution with mean zero is

$$u = \sum_{(m,n) \neq (0,0)} \frac{f_{mn}}{\pi(im - n)} e_{mn}. \quad (1.15)$$

Indeed $|\pi(im - n)| = \pi\sqrt{m^2 + n^2}$, so dividing a rapidly decreasing Fourier sequence by this multiplier still gives a rapidly decreasing sequence. Thus u is smooth and termwise differentiation is justified. The zero mode is the only obstruction, and adding a constant gives all solutions. For instance, if $f = \cos(2\pi x)$, then

$$u = \frac{1}{\pi} \sin(2\pi x), \quad u_{\bar{z}} = f.$$

Similarly, on nonzero modes $G_{\bar{\partial}}$ divides by $2\pi^2(m^2 + n^2)$; the two powers of frequency explain its gain of two Sobolev derivatives. Compactness appears as the discrete frequency lattice and its positive distance from the zero mode.

For a periodic conformal metric $g = \rho^2 g_0$ on this same elliptic curve, dz remains holomorphic and hence Dolbeault harmonic. Its conjugate $d\bar{z}$ is also Dolbeault harmonic: the formula $\bar{\partial}^*(b d\bar{z}) = -2\rho^{-2} b_z$ vanishes for $b = 1$. In top degree a direct star calculation instead gives

$$\bar{\partial}^*(f dz \wedge d\bar{z}) = 2(f\rho^{-2})_z dz.$$

The function $f\rho^{-2}$ must be antiholomorphic and therefore constant on compact E . Thus the harmonic $(1,1)$ -forms are multiples of the new volume form $\omega = (i\rho^2/2)dz \wedge d\bar{z}$. The dimensions of Dolbeault cohomology remain fixed while a metric can change the selected representative. In complex dimension one every Hermitian metric is Kähler; the next section explains both this fact and its failure in higher dimension.

2 The Chern connection: differentiating a Hermitian holomorphic bundle

A holomorphic structure tells us which local sections are holomorphic; a Hermitian metric tells us their lengths. A connection compares sections at nearby points. The Chern connection is the unique connection that respects both pieces of data. For the holomorphic tangent bundle, its torsion will measure precisely the failure of the metric to be Kähler.

2.1 Connections and the two compatibility conditions

A *holomorphic vector bundle* of rank r is a bundle locally identified with $U \times \mathbb{C}^r$, whose linear transition matrices are holomorphic maps to $\mathrm{GL}_r(\mathbb{C})$. Its holomorphic sections have holomorphic coefficient functions in these local frames. A Hermitian bundle metric varies smoothly with the base point; it need not be holomorphic.

Let $E \rightarrow X$ be such a bundle, equipped with a smooth Hermitian metric h , linear in the first argument. A connection is a $\mathbb{C}$ -linear map

$$D : \mathcal{A}^0(X, E) \longrightarrow \mathcal{A}^1(X, E), \quad D(fs) = df \otimes s + fDs.$$

It splits by type as $D = D' + D''$. The holomorphic structure already supplies an operator $\bar{\partial}_E$: in a holomorphic frame it differentiates the coefficient functions by $\bar{\partial}$. This is independent of frame because the transition matrices are holomorphic.

Definition 2.1 (Chern connection). The Chern connection of (E, h) is a connection satisfying

- (i) $D'' = \bar{\partial}_E$;
- (ii) for every real vector field V ,

$$Vh(s, t) = h(D_V s, t) + h(s, D_V t).$$

The second condition is metric compatibility.

Condition (i) says that a holomorphic section has no covariant derivative in antiholomorphic directions. It does not say that a holomorphic section is parallel: its $(1, 0)$ derivative can be nonzero.

2.2 A local formula, with the matrix convention made explicit

Write a holomorphic frame as a row $e = (e_1, \dots, e_r)$ and the coefficients of a section as a column, so the section is es . Define

$$H_{\bar{a}b} = h(e_b, e_a), \quad h(es, et) = \bar{t}^T H s.$$

Thus H is a positive Hermitian matrix. The order of its indices is deliberate: with a first-argument-linear inner product, it gives the usual formula $H^{-1}\partial H$ below. The matrix with entries $h(e_a, e_b)$ would be its transpose and would require the transposed formula.

In this frame write

$$D(es) = e(ds + As), \quad De_b = \sum_a e_a A_{ab}.$$

The condition $D'' = \bar{\partial}_E$ forces A to have type $(1, 0)$. Let $A^\dagger = \bar{A}^T$, including conjugation of differential forms. Applying metric compatibility to the frame sections gives

$$dH = A^\dagger H + HA.$$

Its $(1, 0)$ part is $\partial H = HA$, so necessarily

$$\boxed{A = H^{-1}\partial H, \quad D' = \partial + H^{-1}\partial H, \quad D'' = \bar{\partial}_E.} \quad (2.1)$$

Theorem 2.2 (Existence and uniqueness). *Every Hermitian holomorphic vector bundle has a unique Chern connection.*

Proof. The calculation above proves uniqueness. To prove existence, define A by (2.1) in each holomorphic frame. Since $H^\dagger = H$, we have $A^\dagger = \bar{\partial} H H^{-1}$, and therefore $A^\dagger H + HA = \bar{\partial} H + \partial H = dH$. Both compatibility conditions hold locally.

It remains to check that these local connections agree. If $e' = eB$ with B holomorphic and invertible, then

$$H' = B^\dagger H B, \quad A' = (H')^{-1}\partial H' = B^{-1}AB + B^{-1}\partial B.$$

Here $\partial B^\dagger = 0$. Since $s' = B^{-1}s$ and $dB = \partial B$, the last identity gives $ds' + A's' = B^{-1}(ds + As)$. This is exactly the connection transformation law, so the local formulas glue. $\square$

Example 2.3 (A flat metric in a nonconstant holomorphic frame). On the trivial line bundle over $\mathbb{C}$ take a frame e with $h(e, e) = 1$. Then $A = 0$. Replace it by $e' = e^z e$. Its squared norm is $h' = e^{z+\bar{z}}$ and its connection form is $A' = dz$. Thus a nonzero connection matrix need not indicate curvature: it may record variation of the chosen frame. The curvature in both frames is zero.

2.3 Curvature and Hermitian line bundles

Extend D to bundle-valued forms by the graded Leibniz rule. Its square is multiplication by an endomorphism-valued two-form:

$$D^2 s = F_D s, \quad F_D = dA + A \wedge A.$$

For the Chern connection,

$$\partial(H^{-1}) = -H^{-1}(\partial H)H^{-1}, \quad \partial A = -A \wedge A.$$

Consequently

$$F_D = \bar{\partial}(H^{-1}\partial H), \quad F_D \text{ has type } (1, 1). \quad (2.2)$$

Under a change of frame, $F'_D = B^{-1}F_D B$. Thus the matrix changes, but the curvature tensor and its trace are intrinsic.

For a line bundle write the squared norm of a local holomorphic frame as $h = e^{-\varphi}$, with φ real. Then

$$A = \partial \log h = -\partial \varphi, \quad F_D = -\bar{\partial} \partial \varphi = \partial \bar{\partial} \varphi. \quad (2.3)$$

The order of ∂ and $\bar{\partial}$ determines the sign. In one variable, $F_D = \varphi_{z\bar{z}} dz \wedge d\bar{z}$.

The real closed form

$$c_1(E, h) = \frac{i}{2\pi} \text{Tr } F_D \quad (2.4)$$

represents the image of the first Chern class in real de Rham cohomology. This is the standard Chern–Weil identification; we will use it only to interpret the explicit integrals below. For a line bundle, the independence of the metric is already visible: replacing h by $e^{-\psi}h$, where ψ is globally defined and real, changes its curvature by $\partial \bar{\partial} \psi = d(\bar{\partial} \psi)$, an exact form. For a nonvanishing holomorphic change of frame $e' = be$, the change $\log h' - \log h = \log |b|^2$ has vanishing $\partial \bar{\partial}$, so curvature also agrees across charts.

Example 2.4 (A varying metric on a holomorphically trivial torus bundle). On $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ take the trivial line bundle and a smooth real periodic function φ . The global frame has norm $e^{-\varphi}$, so $A = -\partial \varphi$ and $F_D = \partial \bar{\partial} \varphi$. The curvature can be nonzero pointwise even though its cohomology class is zero. On the square torus, choose $\varphi = a \cos(2\pi x)$. Then

$$A = a\pi \sin(2\pi x) dz, \quad F_D = -a\pi^2 \cos(2\pi x) dz \wedge d\bar{z}, \quad \int_{E_i} c_1(E, h) = 0.$$

The zero integral follows either from the cosine average or from Stokes. Topology constrains the curvature class, while the metric changes its representative.

2.4 The tangent bundle of a Hermitian manifold

Return to a Hermitian metric g on X . In holomorphic coordinates write

$$g(v, w) = \text{Re} \sum_{j,k} G_{j\bar{k}} dz^j(v) \overline{dz^k(w)}, \quad \omega = \frac{i}{2} \sum_{j,k} G_{j\bar{k}} dz^j \wedge d\bar{z}^k,$$

for real tangent vectors v, w . If $g_{\mathbb{C}}$ denotes the complex-bilinear extension of g , give $T^{1,0}X$ the metric

$$h(v, w) = 2g_{\mathbb{C}}(v, \bar{w}).$$

Then $h(\partial_{z^j}, \partial_{z^k}) = G_{j\bar{k}}$. The factor 2 makes the standard tangent frame have squared norm 1 for g_0 ; it is a constant rescaling and does not affect the Chern connection. For the row frame $(\partial_{z^1}, \dots, \partial_{z^n})$, the matrix used in (2.1) is $H_{\bar{a}b} = G_{b\bar{a}}$.

Writing $D_{\partial_{z^i}} \partial_{z^j} = \sum_k \Gamma_{ij}^k \partial_{z^k}$ gives the useful component formula

$$\Gamma_{ij}^k = \sum_{\ell} (G^{-1})_{\ell k} \partial_i G_{j\bar{\ell}}, \quad D_{\partial_{z^i}} \partial_{z^j} = 0. \quad (2.5)$$

Equivalently, $\partial_i G_{j\bar{\ell}} = \sum_k \Gamma_{ij}^k G_{k\bar{\ell}}$. This last identity is a convenient way to check index placement.

Example 2.5 (Every flat complex torus). Translations preserve the holomorphic frame ∂_{z^j} and any constant positive Hermitian matrix G . They descend to $T = \mathbb{C}^n / \Lambda$, where Λ is a lattice of real rank $2n$. Formula (2.5) gives

$$\Gamma_{ij}^k = 0, \quad F_{T^{1,0}T} = 0.$$

The forms dz^j also descend and are parallel. The functions z^j usually do not descend: global invariant differentials do not require global coordinate functions. Later we will see that their constant wedge products give all harmonic forms for this flat metric.

2.5 A single curvature formula for three kinds of curves

On a Riemann surface write $g = \rho(z, \bar{z})|dz|^2$. The holomorphic tangent frame ∂_z has norm squared $h = \rho$, and hence

$$A = (\partial_z \log \rho) dz, \quad F_{TX} = -(\partial_z \partial_{\bar{z}} \log \rho) dz \wedge d\bar{z}. \quad (2.6)$$

The Gaussian curvature formula $K = -\frac{1}{2\rho}(\partial_x^2 + \partial_y^2) \log \rho$ therefore becomes

$$c_1(TX, h) = \frac{K}{2\pi} \omega, \quad \omega = \rho dx \wedge dy. \quad (2.7)$$

Here $\partial_z \partial_{\bar{z}} = \frac{1}{4}(\partial_x^2 + \partial_y^2)$. This is a concrete bridge between complex curvature and the real surface geometry of Lecture 1.

Example 2.6 (The projective line and the sphere). Fix the normalization

$$\omega_{\text{FS}} = \frac{i}{2} \partial \bar{\partial} \log(1 + |z|^2), \quad g_{\text{FS}} = \frac{|dz|^2}{(1 + |z|^2)^2}.$$

Thus $\rho = (1 + |z|^2)^{-2}$, and direct differentiation gives

$$A_{T\mathbb{P}^1} = -\frac{2\bar{z}}{1 + |z|^2} dz, \quad F_{T\mathbb{P}^1} = \frac{2}{(1 + |z|^2)^2} dz \wedge d\bar{z}, \quad c_1(T\mathbb{P}^1) = \frac{2}{\pi} \omega_{\text{FS}}.$$

The omitted point at infinity has measure zero, so polar coordinates give

$$\int_{\mathbb{P}^1} \omega_{\text{FS}} = \int_0^{2\pi} \int_0^\infty \frac{r}{(1 + r^2)^2} dr d\theta = \pi, \quad \int_{\mathbb{P}^1} c_1(T\mathbb{P}^1) = 2.$$

In this convention $K = 4$. The unit round sphere of Lecture 1 has metric $g_{\text{round}} = 4g_{\text{FS}}$ and area 4π . A constant metric rescaling leaves A and F unchanged; it rescales K and area in opposite directions.

Example 2.7 (The upper half-plane and compact hyperbolic quotients). On $\mathbb{H} = \{z = x + iy : y > 0\}$ use

$$g_{\text{hyp}} = \frac{dx^2 + dy^2}{y^2}, \quad \omega_{\text{hyp}} = \frac{i}{2y^2} dz \wedge d\bar{z}.$$

Since $\partial_z y = 1/(2i)$, formula (2.6) yields

$$A = \frac{i}{y} dz, \quad F_{T\mathbb{H}} = -\frac{1}{2y^2} dz \wedge d\bar{z}, \quad c_1(T\mathbb{H}) = -\frac{1}{2\pi} \omega_{\text{hyp}}.$$

For $\gamma(z) = (az + b)/(cz + d)$ in $\text{PSL}_2(\mathbb{R})$, $\text{Im } \gamma(z) = y/|cz + d|^2$ and $\gamma'(z) = 1/(cz + d)^2$. These two factors cancel in $|dz|^2/y^2$, so the metric descends under a freely acting properly discontinuous group Γ . If $X = \Gamma \backslash \mathbb{H}$ is compact of genus $g \geq 2$, Gauss–Bonnet gives

$$\text{Area}(X) = 4\pi(g - 1), \quad \int_X c_1(TX) = 2 - 2g.$$

The local connection form depends on the chart; curvature descends as a global tensor. The signs for the sphere, flat torus, and hyperbolic curve are respectively positive, zero, and negative.

3 The Kähler condition and why it removes torsion

3.1 A pointwise metric, a differential condition

Recall that $\omega(u, v) = g(Ju, v)$ is a real positive $(1, 1)$ -form. Conversely $g(u, v) = \omega(u, Jv)$ recovers g from such a form. In complex dimension n , the orientation determined by the complex coordinates satisfies

$$dV_g = \frac{\omega^n}{n!}. \quad (3.1)$$

At a point with $G = I$, this follows by expanding $\omega = \sum_j dx^j \wedge dy^j$: the $n!$ nonzero terms all give the positive real volume form. Since both sides are tensors, the formula holds in all coordinates.

Definition 3.1. A Hermitian metric is Kähler if its fundamental form is closed: $d\omega = 0$. A complex manifold is called Kähler if it admits such a metric.

The adjective applied to a metric and the adjective applied to a complex manifold answer different questions. A complex torus admits a flat Kähler metric, but it also admits Hermitian metrics that are not Kähler when its complex dimension is at least two.

Because $\partial\omega$ and $\bar{\partial}\omega$ have different types, $d\omega = 0$ is equivalent to the vanishing of each. In coordinates,

$$d\omega = 0 \iff \partial_i G_{j\bar{\ell}} = \partial_j G_{i\bar{\ell}} \text{ for all } i, j, \ell, \quad (3.2)$$

together with the conjugate equations, which follow automatically. Thus Hermitian symmetry is a pointwise condition on G , while Kähler geometry imposes symmetry on its first derivatives.

3.2 Torsion, the Chern connection, and Levi–Civita

The Chern connection on $T^{1,0}X$, together with its conjugate on $T^{0,1}X$, gives a real metric connection on TX preserving J . Its torsion is

$$T(U, V) = D_U V - D_V U - [U, V].$$

In holomorphic coordinates its mixed torsion vanishes, and its $(2, 0)$ part is

$$T(\partial_{z^i}, \partial_{z^j}) = \sum_k (\Gamma_{ij}^k - \Gamma_{ji}^k) \partial_{z^k}. \quad (3.3)$$

The $(0, 2)$ part is the conjugate. Combining (2.5) and (3.2) shows directly that $T = 0$ exactly when $d\omega = 0$.

Theorem 3.2 (Equivalent descriptions of a Kähler metric). *For a Hermitian metric on a complex manifold, the following are equivalent:*

- (i) $d\omega = 0$;
- (ii) the real connection induced by the tangent Chern connection has zero torsion;
- (iii) that connection equals the Levi-Civita connection ∇ ;
- (iv) $\nabla J = 0$;
- (v) $\nabla\omega = 0$.

Proof. The coordinate calculation proves (i) $\Leftrightarrow$ (ii). A real metric connection with zero torsion is uniquely Levi-Civita, so (ii) $\Rightarrow$ (iii). The Chern connection preserves J , giving (iii) $\Rightarrow$ (iv). Since $\nabla g = 0$ and $\omega(U, V) = g(JU, V)$, (iv) and (v) are equivalent. Finally a torsion-free connection satisfies

$$d\omega(U, V, W) = (\nabla_U \omega)(V, W) + (\nabla_V \omega)(W, U) + (\nabla_W \omega)(U, V).$$

Consequently (v) $\Rightarrow$ (i), completing the cycle. $\square$

This theorem supplies the geometric input for the Kähler identities. Parallel transport preserves the decomposition into holomorphic and antiholomorphic directions; differentiating a form by ∇ respects its type. The torsion-free property lets us recover exterior differentiation by alternating the covariant derivative, with no additional torsion term.

3.3 Torus tests: dimension one and dimension two

Every Hermitian metric on a Riemann surface is Kähler: $d\omega$ is a three-form on a real two-manifold and hence vanishes. In particular, on $E_i = \mathbb{C}/(\mathbb{Z} + i\mathbb{Z})$ every metric

$$g_u = e^{2u(x,y)}(dx^2 + dy^2), \quad \omega_u = e^{2u} dx \wedge dy$$

is Kähler for smooth periodic u . Its tangent connection and curvature need not be flat:

$$A = 2\partial u, \quad F = -2\partial\bar{\partial}u, \quad K = -e^{-2u}(u_{xx} + u_{yy}). \quad (3.4)$$

The curvature integrates to zero because the underlying torus has Euler characteristic zero. The one-forms $dz, d\bar{z}$ remain harmonic for these metrics: the real two-dimensional star on one-forms is unchanged by a conformal factor, and these forms are closed and co-closed.

Now let $T^4 = \mathbb{C}^2/(\mathbb{Z} + i\mathbb{Z})^2$ and take $u = \varepsilon \cos(2\pi x^1)$, with $\varepsilon \neq 0$. The conformal metric has fundamental form

$$\omega_u = e^{2u}\omega_0, \quad \omega_0 = dx^1 \wedge dy^1 + dx^2 \wedge dy^2.$$

Here

$$d\omega_u = 2e^{2u}u_{x^1} dx^1 \wedge dx^2 \wedge dy^2 \neq 0. \quad (3.5)$$

For its tangent bundle $H = e^{2u}I$, so

$$A = 2\partial u I, \quad F = -2\partial\bar{\partial}u I, \quad T(\partial_{z^1}, \partial_{z^2}) = 2u_{z^1}\partial_{z^2}.$$

The torsion and $d\omega_u$ detect the same first derivative. This metric is not Kähler; the complex torus still is, because it admits g_0 . Later this example will also show explicitly why the de Rham and Dolbeault harmonic spaces need not agree for a general Hermitian metric.

More generally, if $n \geq 2$ and ω is Kähler, the form $e^{2u}\omega$ is closed only when u is locally constant. Indeed, $du \wedge \omega = 0$ implies $du \wedge \omega^{n-1} = 0$. The pointwise map $\alpha \mapsto \alpha \wedge \omega^{n-1}$ from one-forms to $(2n-1)$ -forms is an isomorphism, as is checked in a unitary basis. Thus $du = 0$.

3.4 Local potentials and their global limitation

Theorem 3.3 (Local Kähler potentials). *A real positive $(1,1)$ -form ω is closed if and only if, near every point, there is a smooth real function φ with*

$$\omega = \frac{i}{2} \partial \bar{\partial} \varphi. \quad (3.6)$$

Proof. A form on the right is closed by $\partial^2 = \bar{\partial}^2 = 0$ and $\partial \bar{\partial} = -\bar{\partial} \partial$. Conversely, shrink to a coordinate ball and use the real Poincaré lemma to write $\omega = d\eta$ with η real. Its type decomposition gives $\bar{\partial}\eta^{0,1} = 0$, since ω has no $(0,2)$ part. The local $\bar{\partial}$ -Poincaré lemma, after shrinking again, gives $\eta^{0,1} = \bar{\partial}f$. Reality gives $\eta^{1,0} = \partial \bar{f}$, and therefore

$$\omega = \partial \bar{\partial} f + \bar{\partial} \partial \bar{f} = \partial \bar{\partial} (f - \bar{f}) = 2i \partial \bar{\partial} \operatorname{Im} f.$$

Take $\varphi = 4 \operatorname{Im} f$. □

In this normalization the metric coefficients are simply $G_{j\bar{k}} = \partial_j \partial_{\bar{k}} \varphi$. Positivity is positive definiteness of this complex Hessian. The potential is not unique: adding the real part of a holomorphic function changes neither the Hessian nor ω . More generally the difference of two real potentials is pluriharmonic, meaning its $\partial \bar{\partial}$ vanishes.

Example 3.4 (Three local potentials). For Euclidean space, take $\varphi = \sum_j |z^j|^2$. For the upper half-plane, take $\varphi = -4 \log y$, since $\partial_z \partial_{\bar{z}}(-4 \log y) = y^{-2}$. For the affine chart of $\mathbb{P}^1$, take $\varphi = \log(1 + |z|^2)$. These potentials produce respectively flat, negative, and positive curvature metrics. Positivity of the complex Hessian is the condition for a positive metric; it does not prescribe the sign of its curvature, which involves further derivatives.

On a flat torus, $\sum_j |z^j|^2$ is not periodic, even though its $\partial \bar{\partial}$ is translation invariant. This is necessary, rather than a poor choice of potential. On any compact Kähler manifold of positive dimension, a global potential would give

$$\omega = d\left(\frac{i}{2} \bar{\partial} \varphi\right), \quad \omega^n = d\left(\frac{i}{2} \bar{\partial} \varphi \wedge \omega^{n-1}\right).$$

Stokes would imply $\int_X \omega^n = 0$, contradicting $\int_X \omega^n = n! \operatorname{Vol}_g(X) > 0$. In particular $[\omega] \neq 0$ in real de Rham cohomology.

3.5 Projective space and the hyperplane line bundle

On $U_0 = \{Z_0 \neq 0\} \subset \mathbb{P}^n$, with $z^j = Z_j/Z_0$, set

$$\varphi_0 = \log(1 + |z|^2), \quad \omega_{\text{FS}} = \frac{i}{2} \partial \bar{\partial} \varphi_0.$$

A direct derivative gives

$$G_{j\bar{k}} = \frac{(1 + |z|^2) \delta_{jk} - \bar{z}^j z^k}{(1 + |z|^2)^2}. \quad (3.7)$$

For a nonzero vector v , its numerator evaluates to

$$(1 + |z|^2)|v|^2 - \left| \sum_j \bar{z}^j v^j \right|^2 \geq |v|^2 > 0$$

by Cauchy–Schwarz. Thus the Hessian is positive. The form is closed by construction. On U_a , use $\varphi_a = \log(\sum_j |Z_j|^2 / |Z_a|^2)$; on an overlap, $\varphi_a - \varphi_b = \log |Z_b/Z_a|^2$ has vanishing $\partial\bar{\partial}$. The local forms therefore define a global Kähler metric on projective space.

The same potential defines the standard metric on the hyperplane line bundle $\mathcal{O}(1)$: in this local frame,

$$h_{\mathcal{O}(1)} = (1 + |z|^2)^{-1}, \quad F_{\mathcal{O}(1)} = \partial\bar{\partial} \log(1 + |z|^2), \quad c_1(\mathcal{O}(1), h) = \frac{\omega_{\text{FS}}}{\pi}.$$

On $\mathbb{P}^1$ this has integral 1, while the tangent curvature computed above has integral 2. Thus the normalized hyperplane class is $[\omega_{\text{FS}}/\pi]$. Keeping this factor explicit will prevent an area normalization from becoming a cohomology normalization error.

3.6 Holomorphic normal coordinates: the bridge to operator identities

Theorem 3.5. *At every point of a Kähler manifold there are holomorphic coordinates centered at that point such that*

$$G_{i\bar{j}}(0) = \delta_{ij}, \quad \partial_k G_{i\bar{j}}(0) = \partial_{\bar{k}} G_{i\bar{j}}(0) = 0.$$

Proof. First make a complex linear coordinate change so $G(0) = I$. Set $C_{ik}^a = \partial_k G_{i\bar{a}}(0)$. The Kähler condition gives $C_{ik}^a = C_{ki}^a$, so the quadratic holomorphic change

$$z^a = w^a - \frac{1}{2} \sum_{i,k} C_{ik}^a w^i w^k$$

has precisely the symmetric second derivative we need. Its differential at zero is the identity, so it gives coordinates near zero. In these coordinates

$$G'_{i\bar{j}} = \sum_{a,b} G_{a\bar{b}}(z(w)) \frac{\partial z^a}{\partial w^i} \frac{\partial \bar{z}^b}{\partial w^j}.$$

Differentiating at zero yields $\partial_k G'_{i\bar{j}}(0) = \partial_k G_{i\bar{j}}(0) - C_{ik}^j = 0$. Hermitian symmetry gives the conjugate derivatives as well. $\square$

These coordinates make the metric Euclidean to first order at a chosen point. They do not make a curved metric Euclidean on a whole neighborhood: curvature survives in second derivatives. For a general Hermitian metric, the antisymmetric part $\partial_k G_{i\bar{a}} - \partial_i G_{k\bar{a}}$ cannot be removed by a holomorphic quadratic coordinate change, because its second derivatives are symmetric. This is the same torsion obstruction encountered above.

For a flat torus the invariant coordinates already have constant metric coefficients. For a curved Kähler manifold the normal-coordinate theorem lets the first-order operator calculation look exactly like that flat model at any selected point. This explains why a torus calculation will be a valid local model for the Kähler identities, provided the geometric hypotheses are carried along. A systematic reference for the connection and Kähler material is Demailly [1].

4 The Kähler identities: the bridge between two complexes

The de Rham and Dolbeault complexes see different kinds of information. The first is defined on the underlying smooth manifold; the second remembers which coordinates are holomorphic. A Hermitian metric gives a Laplacian for each complex. The Kähler condition makes these operators agree, up to a factor of two. This is the mechanism behind the cohomological decomposition.

Throughout this section, X is a Kähler manifold of complex dimension n . Compactness is not required for identities of differential operators. When we use an L^2 energy argument, we will explicitly assume compactness and no boundary, or use compactly supported forms.

4.1 A piece of exterior algebra at one point

Define the Lefschetz operator and its adjoint by

$$L\alpha = \omega \wedge \alpha, \quad \Lambda = L^*.$$

Their bidegrees are $(1, 1)$ and $(-1, -1)$. Here the adjoint is pointwise; since L has order zero, it is also its L^2 adjoint. We use $[A, B] = AB - BA$ when one operator is even, and write $\{A, B\} = AB + BA$ for the anticommutator of two odd operators.

At a point, choose an orthonormal complex coframe $\theta^1, \dots, \theta^n$ of type $(1, 0)$. With our normalization,

$$\theta^j = \frac{dz^j}{\sqrt{2}} \quad \text{in Euclidean coordinates,} \quad \omega = i \sum_j \theta^j \wedge \bar{\theta}^j.$$

The factor $\sqrt{2}$ matters: the real Euclidean metric gives $|dz^j|^2 = 2$. Introduce exterior multiplication and its adjoint:

$$\varepsilon_j = \theta^j \wedge (-), \quad \bar{\varepsilon}_j = \bar{\theta}^j \wedge (-), \quad \iota_j = \varepsilon_j^*, \quad \bar{\iota}_j = \bar{\varepsilon}_j^*.$$

Applying these operators to a wedge of basis covectors gives

$$\begin{aligned} \iota_j \varepsilon_k + \varepsilon_k \iota_j &= \delta_{jk} \text{id}, & \bar{\iota}_j \bar{\varepsilon}_k + \bar{\varepsilon}_k \bar{\iota}_j &= \delta_{jk} \text{id}, \\ \iota_j \bar{\varepsilon}_k + \bar{\varepsilon}_k \iota_j &= 0, & \bar{\iota}_j \varepsilon_k + \varepsilon_k \bar{\iota}_j &= 0. \end{aligned}$$

For example, $\iota_1(\theta^1 \wedge \bar{\theta}^2) = \bar{\theta}^2$, whereas $\bar{\iota}_2(\theta^1 \wedge \bar{\theta}^2) = -\theta^1$. Taking the adjoint reverses the order of a product and conjugates its scalar coefficient, so

$$L = i \sum_j \varepsilon_j \bar{\varepsilon}_j, \quad \Lambda = -i \sum_j \bar{\iota}_j \iota_j.$$

Lemma 4.1. *The pointwise operators satisfy*

$$[\Lambda, \varepsilon_k] = -i\bar{\iota}_k, \quad [\Lambda, \bar{\varepsilon}_k] = i\iota_k.$$

Proof. The summands with $j \neq k$ commute with ε_k and $\bar{\varepsilon}_k$: moving past two distinct contractions changes the sign twice. For $j = k$, suppress the subscript and calculate

$$\begin{aligned} [\bar{\iota}\iota, \varepsilon] &= \bar{\iota}(1 - \varepsilon\iota) - \varepsilon\bar{\iota}\iota = \bar{\iota}, \\ [\bar{\iota}\iota, \bar{\varepsilon}] &= -\bar{\iota}\bar{\varepsilon}\iota - \bar{\varepsilon}\bar{\iota}\iota = -(1 - \bar{\varepsilon}\iota)\iota - \bar{\varepsilon}\bar{\iota}\iota = -\iota. \end{aligned}$$

Multiplication by $-i$ gives the result. □

As a normalization check,

$$\Lambda(\theta^j \wedge \bar{\theta}^j) = -i, \quad \Lambda\omega = n, \quad \Lambda(dz \wedge d\bar{z}) = -2i \quad (n = 1).$$

Everything so far is Hermitian linear algebra. The next step uses derivatives of the metric and therefore needs the Kähler condition.

4.2 From pointwise algebra to differential operators

Theorem 4.2 (Kähler identities). *With $\omega_0 = \frac{i}{2} \sum_j dz^j \wedge d\bar{z}^j$ and the inner product linear in the first variable,*

$$[\Lambda, \partial] = i\bar{\partial}^*, \quad [\Lambda, \bar{\partial}] = -i\partial^*, \quad (4.1)$$

$$[L, \partial^*] = i\bar{\partial}, \quad [L, \bar{\partial}^*] = -i\partial. \quad (4.2)$$

Proof. Fix $x \in X$. The Levi-Civita connection ∇ preserves J , the metric, and ω because the metric is Kähler. Choose a local orthonormal $(1, 0)$ -form coframe θ^j for the metric on complexified forms, and let e_j be its dual vector frame, such that $\nabla e_j(x) = \nabla \theta^j(x) = 0$. In normal holomorphic coordinates, $\theta^j(x) = dz^j / \sqrt{2}$ and $e_j(x) = \sqrt{2} \partial_{z^j}$. This normalization concerns the form metric. The Chern bundle metric $h = 2g_{\mathbb{C}}(-, \overline{-})$ gives $h(e_j, e_j) = 2$ at x . The coframe can be obtained by orthonormalizing the coordinate coframe. Its normalization need not be holomorphic away from x .

The torsion-free identity expressing d as alternation of ∇ , followed by separation of types, gives at x

$$\partial = \sum_j \varepsilon_j \nabla_{e_j}, \quad \bar{\partial} = \sum_j \bar{\varepsilon}_j \nabla_{\bar{e}_j}.$$

Integration by parts and the vanishing connection coefficients give

$$\partial^* = - \sum_j \iota_j \nabla_{\bar{e}_j}, \quad \bar{\partial}^* = - \sum_j \bar{\iota}_j \nabla_{e_j}.$$

These are identities at the chosen point, including the order-zero terms: those terms vanish in this frame at x . Since $\nabla \Lambda = 0$, the lemma now gives

$$\begin{aligned} [\Lambda, \partial] &= \sum_j [\Lambda, \varepsilon_j] \nabla_{e_j} = -i \sum_j \bar{\iota}_j \nabla_{e_j} = i\bar{\partial}^*, \\ [\Lambda, \bar{\partial}] &= \sum_j [\Lambda, \bar{\varepsilon}_j] \nabla_{\bar{e}_j} = i \sum_j \iota_j \nabla_{\bar{e}_j} = -i\partial^*. \end{aligned}$$

The point x was arbitrary. Finally, take formal adjoints. For example,

$$[\Lambda, \partial]^* = \partial^* L - L \partial^* = -[L, \partial^*] = -i\bar{\partial},$$

which gives the first identity in (4.2); the other follows in the same way. $\square$

Remark 4.3 (Where the geometry enters). A Hermitian metric always permits a unitary basis at a point. It need not permit holomorphic coordinates that also remove all first derivatives of the metric there. Equivalently, its Chern connection can have torsion. Using a torsionful connection to express d introduces extra terms; using Levi-Civita need not preserve type. Thus the pointwise lemma remains true for every Hermitian metric, while the differential identities acquire torsion corrections. This distinction prevents a common incorrect proof of the Kähler identities on an arbitrary complex manifold.

Example 4.4 (Check both signs on the complex line). For $g = dx^2 + dy^2$ and $z = x + iy$, integration by parts gives

$$\partial^*(f dz) = -2f_{\bar{z}}, \quad \bar{\partial}^*(f d\bar{z}) = -2f_z.$$

Consequently,

$$\begin{aligned} [\Lambda, \partial](f d\bar{z}) &= \Lambda(f_z dz \wedge d\bar{z}) = -2if_z = i\bar{\partial}^*(f d\bar{z}), \\ [\Lambda, \bar{\partial}](f dz) &= \Lambda(-f_{\bar{z}} dz \wedge d\bar{z}) = 2if_{\bar{z}} = -i\partial^*(f dz). \end{aligned}$$

The minus sign in $\bar{\partial}^*(f dz) = -f_{\bar{z}} dz \wedge d\bar{z}$ comes from exchanging the two one-forms. The same calculations hold locally on a flat complex torus.

4.3 Deriving the equality of the three Laplacians

Write

$$\Delta_\partial = \partial\partial^* + \partial^*\partial, \quad \Delta_{\bar{\partial}} = \bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial}, \quad \Delta_d = dd^* + d^*d.$$

The two mixed terms in the expansion of Δ_d are the potential obstruction to an operator that preserves type.

Proposition 4.5. *On a Kähler manifold,*

$$\{\partial, \bar{\partial}^*\} = 0, \quad \{\bar{\partial}, \partial^*\} = 0, \quad \boxed{\Delta_d = 2\Delta_\partial = 2\Delta_{\bar{\partial}}}.$$

Proof. First use $\bar{\partial}^* = -i[\Lambda, \partial]$ and $\partial^2 = 0$:

$$\partial\bar{\partial}^* + \bar{\partial}^*\partial = -i(\partial\Lambda\partial - \partial^2\Lambda + \Lambda\partial^2 - \partial\Lambda\partial) = 0.$$

The other anticommutator follows from $\partial^* = i[\Lambda, \bar{\partial}]$ and $\bar{\partial}^2 = 0$. Next, using $\partial\bar{\partial} = -\bar{\partial}\partial$, compute

$$\begin{aligned} \Delta_\partial &= i(\partial[\Lambda, \bar{\partial}] + [\Lambda, \bar{\partial}]\partial) \\ &= i(\partial\Lambda\bar{\partial} - \partial\bar{\partial}\Lambda + \Lambda\bar{\partial}\partial - \bar{\partial}\Lambda\partial) \\ &= i([\partial, \Lambda]\bar{\partial} + \bar{\partial}[\partial, \Lambda]) \\ &= \bar{\partial}^*\bar{\partial} + \bar{\partial}\bar{\partial}^* = \Delta_{\bar{\partial}}. \end{aligned}$$

Finally $d = \partial + \bar{\partial}$ and $d^* = \partial^* + \bar{\partial}^*$, so

$$\Delta_d = \Delta_\partial + \Delta_{\bar{\partial}} + \{\partial, \bar{\partial}^*\} + \{\bar{\partial}, \partial^*\} = 2\Delta_{\bar{\partial}}.$$

□

Example 4.6 (Direct normal-coordinate calculation of the factor two). Fix a point x and choose Kähler holomorphic normal coordinates there. At x , write ε_j for exterior multiplication by dz^j and ι_j for contraction with ∂_{z_j} , with barred notation for the conjugate operators. Since $G_{j\bar{k}}(x) = \delta_{jk}$ and all first coordinate derivatives of G vanish at x , the principal parts of the operators are the flat ones:

$$\partial = \sum_j \varepsilon_j \nabla_j, \quad \bar{\partial} = \sum_j \bar{\varepsilon}_j \nabla_{\bar{j}}, \quad \partial^* = -2 \sum_j \iota_j \nabla_{\bar{j}}, \quad \bar{\partial}^* = -2 \sum_j \bar{\iota}_j \nabla_j.$$

At the chosen point, the lower-order connection terms vanish. Using $\varepsilon_j \iota_k + \iota_k \varepsilon_j = \delta_{jk}$ and the analogous barred identity gives

$$\begin{aligned} \Delta_\partial &= \partial\partial^* + \partial^*\partial = -2 \sum_j \nabla_j \nabla_{\bar{j}}, \\ \Delta_{\bar{\partial}} &= \bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial} = -2 \sum_j \nabla_j \nabla_{\bar{j}}. \end{aligned}$$

Thus $\Delta_\partial = \Delta_{\bar{\partial}}$ at x . Since $\partial_j = \frac{1}{2}(\partial_{x_j} - i\partial_{y_j})$ and $\partial_{\bar{j}} = \frac{1}{2}(\partial_{x_j} + i\partial_{y_j})$,

$$-4 \sum_j \nabla_j \nabla_{\bar{j}} = - \sum_j (\nabla_{x_j}^2 + \nabla_{y_j}^2)$$

is the real de Rham Laplacian at x . Therefore

$$\Delta_d = \Delta_\partial + \Delta_{\bar{\partial}} = 2\Delta_{\bar{\partial}}.$$

This calculation is pointwise: curvature enters when commuting covariant second derivatives, while the Kähler normal-coordinate conditions remove the first-order metric terms that would otherwise obscure the factor of two.

On a compact manifold without boundary, the energy identity makes the consequence especially concrete:

$$\begin{aligned}\Delta_d \alpha = 0 &\iff \Delta_{\bar{\partial}} \alpha = 0 \iff \Delta_{\partial} \alpha = 0, \\ \Delta_d \alpha = 0 &\iff \partial \alpha = \bar{\partial} \alpha = \partial^* \alpha = \bar{\partial}^* \alpha = 0.\end{aligned}\tag{4.3}$$

For instance, the $\bar{\partial}$ statement follows from $(\Delta_{\bar{\partial}} \alpha, \alpha) = \|\bar{\partial} \alpha\|^2 + \|\bar{\partial}^* \alpha\|^2$. On a noncompact manifold, the equality of the differential operators still holds. The implication from a zero Laplacian to these four first-order equations requires an appropriate global integration argument.

Example 4.7 (The factor of two on functions and on Fourier modes). On the Euclidean complex line,

$$\Delta_d f = -(f_{xx} + f_{yy}) = -4f_{z\bar{z}}, \quad \Delta_{\bar{\partial}} f = -2f_{z\bar{z}}.$$

On $E = \mathbb{C}/(\mathbb{Z} + i\mathbb{Z})$, put $e_{m,n}(x, y) = e^{2\pi i(mx + ny)}$. Then

$$\Delta_d e_{m,n} = 4\pi^2(m^2 + n^2)e_{m,n}, \quad \Delta_{\bar{\partial}} e_{m,n} = 2\pi^2(m^2 + n^2)e_{m,n}.$$

The same eigenvalues apply after wedging with any constant form of fixed type. Every nonzero frequency has positive energy; frequency $(0, 0)$ is the entire harmonic subspace. This agrees with the direct Dolbeault computation in Example 1.15.

5 Hodge decomposition and its geometry

We now assume that X is compact, without boundary, and Kähler, of complex dimension n . The analytic results needed below are the de Rham and Dolbeault Hodge theorems: each cohomology class has a unique harmonic representative. Their proofs use ellipticity, regularity, and closed range, as explained in Lecture 1 and in the Dolbeault discussion above. The new input is the equality of Laplacians.

5.1 From harmonic forms to a decomposition of cohomology

Let

$$\mathcal{H}_d^k(X) = \ker \Delta_d|_{\mathcal{A}^k(X; \mathbb{C})}, \quad \mathcal{H}_{\bar{\partial}}^{p,q}(X) = \ker \Delta_{\bar{\partial}}|_{\mathcal{A}^{p,q}(X)}.$$

Because $\bar{\partial}$ raises q by one and $\bar{\partial}^*$ lowers it by one, $\Delta_{\bar{\partial}}$ preserves bidegree. Hence $\Delta_d = 2\Delta_{\bar{\partial}}$ also preserves bidegree. If $\alpha = \sum_{p+q=k} \alpha^{p,q}$, then

$$\Delta_d \alpha = 0 \iff \Delta_d \alpha^{p,q} = 0 \text{ for every } (p, q).$$

The pointwise orthogonality of distinct types therefore gives

$$\mathcal{H}_d^k(X) = \bigoplus_{p+q=k} \mathcal{H}_{\bar{\partial}}^{p,q}(X).\tag{5.1}$$

Theorem 5.1 (Hodge decomposition for a compact Kähler manifold). *Define*

$$H^{p,q}(X) = \{[\alpha]_{\text{dR}} : \alpha \in \mathcal{H}_{\bar{\partial}}^{p,q}(X)\} \subset H_{\text{dR}}^{p+q}(X; \mathbb{C}).$$

Then

$$\boxed{H_{\text{dR}}^k(X; \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X)}, \quad H_{\bar{\partial}}^{p,q}(X) \simeq H^{p,q}(X).$$

Consequently, with $h^{p,q} = \dim_{\mathbb{C}} H_{\bar{\partial}}^{p,q}(X)$,

$$b_k(X) = \sum_{p+q=k} h^{p,q}(X).$$

Proof. The de Rham Hodge theorem identifies $H_{\text{dR}}^k(X; \mathbb{C})$ with $\mathcal{H}_d^k(X)$. Apply (5.1) to its unique harmonic representative. Every summand is d -closed, so each defines a cohomology class. If a sum of these classes is zero, the sum of the harmonic forms is exact and harmonic, hence zero by the energy argument of Lecture 1. The type components must each be zero. This proves both spanning and directness. Finally, Dolbeault Hodge theory identifies $H_{\bar{\partial}}^{p,q}(X)$ with $\mathcal{H}_{\bar{\partial}}^{p,q}(X)$, proving the second assertion. Taking dimensions gives the formula for b_k . $\square$

Remark 5.2 (What the map actually does). For a Dolbeault class, the construction is

$$[\alpha]_{\bar{\partial}} \longmapsto \text{its } \bar{\partial}\text{-harmonic representative } h \longmapsto [h]_{\text{dR}}.$$

An arbitrary $\bar{\partial}$ -closed form need not be d -closed, even on a compact Kähler manifold. For example, on the square elliptic curve, $\alpha = e^{2\pi i x} d\bar{z}$ is automatically $\bar{\partial}$ -closed, but $\partial\alpha = \pi i e^{2\pi i x} dz \wedge d\bar{z} \neq 0$. Its Dolbeault class is zero, since $\alpha = \bar{\partial}(e^{2\pi i x}/(\pi i))$. The displayed construction sends it to zero; the expression $[\alpha]_{\text{dR}}$ itself is undefined.

5.2 What depends on the metric?

The harmonic representative depends on the chosen metric, as we saw in Lecture 1. The subspace $H^{p,q}(X)$ of de Rham cohomology is independent of which Kähler metric is used. The following characterization separates these statements.

Proposition 5.3. *A class belongs to $H^{p,q}(X)$ if and only if it admits a d -closed representative of pure type (p, q) . Moreover, the isomorphism $H_{\bar{\partial}}^{p,q}(X) \rightarrow H^{p,q}(X)$ above is independent of the Kähler metric.*

Proof. One implication is immediate. Conversely, suppose $d\alpha = 0$ and α has type (p, q) . Its harmonic projection $P_d\alpha$ has the same type, since the harmonic space splits orthogonally by type. The de Rham Hodge theorem says that $\alpha - P_d\alpha$ is exact. Thus its class lies in $H^{p,q}(X)$.

For the assertion about maps, let h_1, h_2 be the harmonic representatives of a fixed Dolbeault class for two Kähler metrics. They are both d -closed, and $h_1 - h_2 = \bar{\partial}\gamma$. Use the first metric for the following orthogonality argument. Every d -harmonic form η splits into $\bar{\partial}$ -harmonic type components, so

$$(h_1 - h_2, \eta) = (\bar{\partial}\gamma, \eta^{p,q}) = (\gamma, \bar{\partial}^*\eta^{p,q}) = 0.$$

The closed form $h_1 - h_2$ has zero harmonic projection. Hodge decomposition therefore makes it d -exact. The two constructions give the same de Rham class. $\square$

This also proves that cup product respects type:

$$H^{p,q}(X) \smile H^{r,s}(X) \subset H^{p+r, q+s}(X).$$

Indeed, the wedge of two closed pure-type representatives is closed and of the indicated type. This assertion does not say that their wedge is harmonic. We will see an explicit failure on an elliptic curve below.

Corollary 5.4. *Every holomorphic p -form on a compact Kähler manifold is d -closed.*

Proof. For a holomorphic $(p, 0)$ -form α , one has $\bar{\partial}\alpha = 0$ and $\bar{\partial}^*\alpha = 0$, the latter for degree reasons. Hence $\Delta_{\bar{\partial}}\alpha = 0$, and the equality of Laplacians makes α d -harmonic. Compactness and the energy identity imply $d\alpha = 0$. $\square$

Both hypotheses have a role. On noncompact Euclidean $\mathbb{C}^2$, the form $z^1 dz^2$ is holomorphic and is annihilated by Δ_d , yet $d(z^1 dz^2) = dz^1 \wedge dz^2 \neq 0$. Its lack of decay prevents the global energy argument. Later we will see a compact Hermitian example where the equality of Laplacians itself fails.

5.3 Conjugation, Hodge star, and Poincaré duality

Our Hodge star is the complex-linear extension of the real Hodge star, and the metric inner product is

$$(\alpha, \beta)_{L^2} = \int_X \alpha \wedge * \bar{\beta}.$$

The three operations below have different effects:

operation	type of the image of (p, q)	linearity
$\alpha \mapsto \bar{\alpha}$	(q, p)	conjugate-linear
$\alpha \mapsto *\alpha$	$(n - q, n - p)$	complex-linear
$\alpha \mapsto *\bar{\alpha}$	$(n - p, n - q)$	conjugate-linear

For the star's type, observe that $\alpha \wedge *\bar{\alpha}$ must have type (n, n) . Checking basis wedges in a unitary coframe proves the stated map for all forms. Both star and conjugation commute with Δ_d , so all three operations preserve harmonicity. Consequently

$$h^{p,q} = h^{q,p} = h^{n-p,n-q} = h^{n-q,n-p}.$$

Conjugation gives the first equality; integration duality gives the complementary-bidegree equality.

Proposition 5.5 (Poincaré duality in Hodge bidegrees). *The complex-bilinear integration pairing*

$$H^{p,q}(X) \times H^{n-p,n-q}(X) \longrightarrow \mathbb{C}, \quad (a, b) \longmapsto \int_X a \wedge b$$

is nondegenerate. Under Dolbeault cohomology it is the pairing $H^q(X, \Omega_X^p) \times H^{n-q}(X, \Omega_X^{n-p}) \rightarrow \mathbb{C}$.

Proof. Stokes' theorem makes the integral independent of closed representatives. Given a nonzero class, take its harmonic representative α and set $\beta = *\bar{\alpha}$. This is harmonic of type $(n - p, n - q)$ and

$$\int_X \alpha \wedge \beta = \int_X \alpha \wedge *\bar{\alpha} = \|\alpha\|_{L^2}^2 > 0.$$

The same argument in the other degree proves nondegeneracy on both sides. $\square$

The integration pairing is bilinear. Positivity in this proof comes from the conjugate-linear choice of the second argument $*\bar{\alpha}$; it does not turn the pairing itself into a Hermitian inner product. For Dolbeault representatives, invariance also follows directly from Stokes: $\int_X \bar{\partial}\gamma \wedge \beta = 0$ when $\bar{\partial}\beta = 0$ and the total type is (n, n) . This is the differential-form realization of Serre duality.

Corollary 5.6 (An immediate topological restriction). *Every odd Betti number of a compact Kähler manifold is even.*

Proof. If $p + q$ is odd, $p \neq q$. Conjugation pairs the distinct spaces $H^{p,q}$ and $H^{q,p}$, which have equal dimension. Their contributions to b_{p+q} therefore occur in equal pairs. $\square$

6 Worked examples and the Kähler hypothesis

6.1 The elliptic curve: periods determine the Hodge lines

Let

$$E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z}), \quad \text{Im } \tau > 0.$$

Write $z = x + \tau y$ with x, y real modulo one. The coordinate z is not a global function, but $dz = dx + \tau dy$ and $d\bar{z} = dx + \bar{\tau} dy$ are global one-forms. For the flat metric $|dz|^2$, their coefficients are constant, so

$$H^{1,0}(E_\tau) = \mathbb{C}[dz], \quad H^{0,1}(E_\tau) = \mathbb{C}[d\bar{z}], \quad H^1(E_\tau; \mathbb{C}) = \mathbb{C}[dz] \oplus \mathbb{C}[d\bar{z}].$$

Let a and b be the coordinate circles $y = 0$ and $x = 0$, respectively, oriented by increasing x and y . Their periods are

$$\begin{array}{c|cc} & a & b \\ \hline dz & 1 & \tau \\ d\bar{z} & 1 & \bar{\tau} \end{array}.$$

The determinant $\bar{\tau} - \tau \neq 0$ proves directly that the two classes form a basis. Solving the two linear equations gives

$$\begin{aligned} [dy] &= \frac{[dz] - [d\bar{z}]}{\tau - \bar{\tau}}, \\ [dx] &= \frac{\tau[d\bar{z}] - \bar{\tau}[dz]}{\tau - \bar{\tau}}. \end{aligned}$$

Thus the real vector space $H^1(\mathbb{R}^2/\mathbb{Z}^2; \mathbb{R})$ can stay fixed while its two complex Hodge lines vary with τ . The splitting remembers the complex structure. It does not arise from the topology of a smooth torus alone.

With the complex orientation,

$$dz \wedge d\bar{z} = -2i \operatorname{Im} \tau \, dx \wedge dy, \quad \omega = \operatorname{Im} \tau \, dx \wedge dy, \quad \operatorname{Area}(E_\tau) = \operatorname{Im} \tau.$$

In particular,

$$\int_{E_\tau} dz \wedge d\bar{z} = -2i \operatorname{Im} \tau, \quad \eta = \frac{i}{2 \operatorname{Im} \tau} dz \wedge d\bar{z} = dx \wedge dy, \quad \int_{E_\tau} \eta = 1.$$

This explicitly computes the perfect pairing $H^{1,0} \times H^{0,1}$ and identifies the integral generator of $H^2(E_\tau; \mathbb{Z})$. On one-forms, $*dz = -idz$ and $*d\bar{z} = id\bar{z}$, so

$$\int_{E_\tau} dz \wedge *d\bar{z} = 2 \operatorname{Im} \tau > 0.$$

6.2 Flat complex tori: harmonic forms and the analytic inverse

Let $T = \mathbb{C}^n/\Gamma$, where Γ is a lattice of real rank $2n$, and choose any translation-invariant Hermitian metric. Its fundamental form is constant and hence closed, so T is Kähler. The invariant holomorphic tangent frame has constant metric matrix; its Chern connection and curvature vanish.

All forms can be expanded in invariant wedges $dz^I \wedge d\bar{z}^J$, whose coefficients are smooth periodic functions. A character of the real torus is $e_\nu(x) = e^{2\pi i \nu(x)}$, with $\nu \in \Gamma^*$ the real dual lattice. Constant-coefficient differentiation gives

$$\Delta_d(e_\nu \alpha_0) = 4\pi^2 |\nu|_{g^{-1}}^2 e_\nu \alpha_0, \quad \Delta_{\bar{d}}(e_\nu \alpha_0) = 2\pi^2 |\nu|_{g^{-1}}^2 e_\nu \alpha_0,$$

where α_0 is any invariant form. The norm is the dual metric norm on real covectors. Therefore

$$\mathcal{H}^{p,q}(T) = \operatorname{span}_{\mathbb{C}}\{dz^I \wedge d\bar{z}^J : |I| = p, |J| = q\}, \quad h^{p,q}(T) = \binom{n}{p} \binom{n}{q}. \quad (6.1)$$

Every harmonic form is translation-invariant, and all invariant forms are harmonic. The de Rham identity $b_k(T) = \binom{2n}{k}$ is recovered from $\sum_{p+q=k} \binom{n}{p} \binom{n}{q} = \binom{2n}{k}$.

Example 6.1 (A Green operator that can be written down). On the square elliptic curve, write a smooth function as $f = \sum_{m,n} \hat{f}_{m,n} e_{m,n}$. The equation $\Delta_d u = f$ has a solution precisely when $\hat{f}_{0,0} = \int_E f dx dy = 0$. Necessity follows either by integration or by looking at the zero frequency. For sufficiency, the unique solution with mean zero is

$$u = \sum_{(m,n) \neq (0,0)} \frac{\hat{f}_{m,n}}{4\pi^2(m^2 + n^2)} e_{m,n}.$$

For $\Delta_{\bar{\partial}} u = f$, replace $4\pi^2$ by $2\pi^2$. Smoothness of f means its Fourier coefficients decrease faster than any power; division by a quadratic frequency preserves rapid decay and gains two derivatives in Sobolev norms. Thus this is a smooth solution, not merely a formal series.

For instance, if $f = \cos(2\pi x)$, then

$$\Delta_d \left(\frac{\cos(2\pi x)}{4\pi^2} \right) = f, \quad \Delta_{\bar{\partial}} \left(\frac{\cos(2\pi x)}{2\pi^2} \right) = f.$$

The same construction applies coefficient by coefficient to differential forms. The harmonic projection is the average of each coefficient.

This calculation exhibits the analytic structure used abstractly in Hodge theory. The kernel is finite-dimensional because it consists of zero frequencies. The positive spectrum is bounded away from zero because the dual lattice is discrete. Division on nonzero frequencies defines an inverse on the orthogonal complement of the kernel. Finally, decay estimates give regularity and the boundedness of that inverse between Sobolev spaces. For an arbitrary compact manifold, elliptic estimates and compactness replace the explicit Fourier diagonalization.

For a complex two-dimensional torus the Hodge numbers are

$$\begin{array}{c|ccc} q \backslash p & 0 & 1 & 2 \\ \hline 2 & 1 & 2 & 1 \\ 1 & 2 & 4 & 2 \\ 0 & 1 & 2 & 1 \end{array} \quad \begin{array}{ccccc} & & 1 & & \\ & & 2 & & 2 \\ 1 & & 4 & & 1 \\ & & 2 & & 2 \\ & & 1 & & \end{array}.$$

In particular $b_2 = 6$. Its four-dimensional space $H^{1,1}$ has basis $[dz^i \wedge d\bar{z}^j]$ for $1 \leq i, j \leq 2$; its remaining two degree-two directions are $[dz^1 \wedge dz^2]$ and its complex conjugate. This is the complex-geometric refinement of the six constant two-forms on the real four-torus studied in Lecture 1.

6.3 Curves and the conformal invariance of harmonic one-forms

Every Hermitian metric on a complex curve is Kähler, because $d\omega$ would have real degree three. If C is a compact connected curve of genus g , then $b_0 = b_2 = 1$ and $b_1 = 2g$. Hodge symmetry gives

$$h^{0,0} = h^{1,1} = 1, \quad h^{1,0} = h^{0,1} = g, \quad \begin{array}{ccc} & 1 & \\ g & & g \\ & 1 & \end{array}.$$

The harmonic $(1,0)$ -forms are exactly the holomorphic differentials, and the harmonic $(0,1)$ -forms are their conjugates. In a holomorphic coordinate $z = x + iy$ with metric $\rho^2 |dz|^2$,

$$*dz = -idz, \quad *d\bar{z} = id\bar{z}.$$

The star on one-forms is independent of ρ . Thus the harmonic one-forms are unchanged by conformal changes of metric, whereas the harmonic representative of the degree-two generator is $\omega / \int_C \omega$ and usually changes.

Example 6.2 (Harmonic forms whose wedge is not harmonic). On the square elliptic curve choose $g_u = e^{2u(x,y)}(dx^2 + dy^2)$ with a smooth, nonconstant periodic function u . It is still Kähler. The forms dz and $d\bar{z}$ are harmonic, but

$$dz \wedge d\bar{z} = -2i \, dx \wedge dy = -2ie^{-2u}\omega_u$$

is not harmonic: a top-degree form on a compact connected oriented surface is harmonic exactly when it is a constant multiple of the volume form. Its harmonic representative is

$$-\frac{2i}{\int_E e^{2u} dx dy} \omega_u.$$

Both have integral $-2i$, hence represent the same class. This gives a concrete distinction between multiplication in cohomology and multiplication of the chosen harmonic representatives.

Example 6.3 (A compact quotient of the upper half-plane). Let $C = \Gamma \backslash \mathbb{H}$ be compact, with Γ torsion-free, acting by real Möbius transformations, and equip it with

$$g_{\text{hyp}} = y^{-2}(dx^2 + dy^2), \quad \omega_{\text{hyp}} = y^{-2} dx \wedge dy.$$

A holomorphic one-form upstairs has the form $f(z)dz$. It descends precisely when

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^2 f(z) \quad \text{for every } \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma.$$

Indeed, $d(\gamma z) = dz/(cz+d)^2$. Such a form is closed because $df \wedge dz = f'(z)dz \wedge dz = 0$, and it is coclosed because $*(fdz) = -idfz$. Its L^2 density is particularly simple:

$$|fdz|_{g_{\text{hyp}}}^2 = 2y^2|f|^2, \quad |fdz|_{g_{\text{hyp}}}^2 \omega_{\text{hyp}} = 2|f|^2 dx dy.$$

If the genus is $g \geq 2$, these descended forms form a complex vector space of dimension g . The normalized harmonic two-form is $\omega_{\text{hyp}}/(4\pi(g-1))$, by Gauss–Bonnet. Compactness is essential here: quotients with cusps require separate conditions at infinity before one can identify an L^2 harmonic space with a chosen cohomology theory.

6.4 Projective space and the sphere normalization

On $\mathbb{P}^n$, use

$$\omega_{\text{FS}} = \frac{i}{2} \partial \bar{\partial} \log(1 + |z^1|^2 + \cdots + |z^n|^2), \quad \eta = \frac{\omega_{\text{FS}}}{\pi}.$$

The form η represents the hyperplane class $c_1(\mathcal{O}(1))$. It is parallel, as is every power η^j , and hence these powers are harmonic. The cellular computation of projective-space cohomology gives one generator in each degree $0, 2, \dots, 2n$ and none in odd degrees. Since η^j has type (j, j) , Hodge decomposition now determines every Hodge number:

$$h^{p,q}(\mathbb{P}^n) = \begin{cases} 1, & p = q, \\ 0, & p \neq q, \end{cases} \quad H^{j,j}(\mathbb{P}^n) = \mathbb{C}[\eta^j], \quad \int_{\mathbb{P}^n} \eta^n = 1.$$

For example, the Hodge diamond of $\mathbb{P}^2$ is

$$\begin{array}{ccccc} & & 1 & & \\ & 0 & & 0 & \\ 0 & & 1 & & 0 \\ & 0 & & 0 & \\ & & 1 & & \end{array}.$$

The middle pairing is explicit: $\int_{\mathbb{P}^2} \eta \wedge \eta = 1$.

On $\mathbb{P}^1$, the affine coordinate formula becomes

$$g_{\text{FS}} = \frac{dx^2 + dy^2}{(1 + |z|^2)^2}, \quad \omega_{\text{FS}} = \frac{dx \wedge dy}{(1 + |z|^2)^2}.$$

Direct polar integration gives

$$\int_{\mathbb{P}^1} \omega_{\text{FS}} = 2\pi \int_0^\infty \frac{r}{(1 + r^2)^2} dr = \pi.$$

Thus the unit round sphere of Lecture 1 has metric $4g_{\text{FS}}$ and area 4π . Its normalized harmonic volume form is still ω_{FS}/π . There are no harmonic one-forms, and therefore no nonzero holomorphic one-forms. Constant rescaling changes the Laplacian's positive eigenvalues and the unnormalized volume, while preserving this cohomological conclusion.

6.5 Two failures that locate the Kähler hypothesis

Example 6.4 (A complex torus with a Hermitian metric that is not Kähler). On $T = \mathbb{C}^2/(\mathbb{Z}^2 + i\mathbb{Z}^2)$, start with the flat metric g_0 and put

$$u = \varepsilon \cos(2\pi x^1), \quad \varepsilon \neq 0, \quad g_u = e^{2u} g_0, \quad \omega_u = e^{2u} \omega_0.$$

Then

$$d\omega_u = 2e^{2u} du \wedge \omega_0 = 2e^{2u} u_{x^1} dx^1 \wedge dx^2 \wedge dy^2$$

is not identically zero. The underlying complex torus does admit the flat Kähler metric; this particular metric is not Kähler.

The global holomorphic form $\alpha = dz^1$ is $\bar{\partial}$ -harmonic for every Hermitian metric: $\bar{\partial}\alpha = 0$ and $\bar{\partial}^*\alpha = 0$ by its type. But it is not d -harmonic for g_u . In real dimension four the conformal star on k -forms is $*_u = e^{(4-2k)u} *_0$. Substituting it into the formula for the codifferential gives, on one-forms,

$$d_u^* \alpha = e^{-2u} (d_0^* \alpha - 2 \iota_{\nabla_0 u} \alpha).$$

For our constant form this reads

$$d_u^* dz^1 = -2e^{-2u} u_{x^1} \neq 0.$$

The compact energy identity implies $\Delta_{d,u} dz^1 \neq 0$, whereas $\Delta_{\bar{\partial},u} dz^1 = 0$. This explicitly disproves the equality of Laplacians for this Hermitian metric and shows where torsion corrections must enter the commutator calculation.

Example 6.5 (A compact complex manifold with no Kähler metric). Consider the Hopf surface

$$S = (\mathbb{C}^2 \setminus \{0\}) / \langle z \mapsto 2z \rangle.$$

Write $z = r\xi$ with $\xi \in S^3$. Multiplication by 2 shifts $\log r$ by $\log 2$ and fixes ξ , so S is diffeomorphic to $S^3 \times S^1$. It is compact and has $b_1 = 1$, $b_2 = 0$. Since b_1 is odd, it cannot admit a Kähler metric.

There is a second proof using only degree two. If ω were a Kähler form, $H^2(S; \mathbb{R}) = 0$ would give $\omega = d\theta$. Then

$$\int_S \omega^2 = \int_S d(\theta \wedge \omega) = 0,$$

contradicting $\omega^2 = 2 \text{vol}_g > 0$. A Hermitian metric certainly exists: $|z|^{-2} g_0$ is invariant under $z \mapsto 2z$ and descends. Its fundamental form is not closed. This example distinguishes the existence of a complex structure, the existence of a Hermitian metric, and the existence of a Kähler metric.

The progression is now visible in the examples. A metric selects harmonic representatives. A complex structure supplies types and the Dolbeault complex. The Kähler condition makes the two Laplacians compatible, and compactness turns that operator identity into the Hodge decomposition of singular cohomology with complex coefficients.

7 The $\partial\bar{\partial}$ lemma and the Frölicher spectral sequence

The Kähler identities give a particularly efficient route from complex geometry to ordinary de Rham cohomology. This section records two cohomological ways to package the same phenomenon: the $\partial\bar{\partial}$ lemma and degeneration of the Dolbeault-to-de Rham spectral sequence.

7.1 The $\partial\bar{\partial}$ lemma

Let X be a complex manifold. Since $d = \partial + \bar{\partial}$, a form can be closed for d while its individual type components need not be closed for both operators. The following lemma is the mechanism that removes this ambiguity on a compact Kähler manifold.

Theorem 7.1 ($\partial\bar{\partial}$ lemma). *Let X be a compact Kähler manifold. If a pure-type form α is d -closed and is either ∂ -exact or $\bar{\partial}$ -exact, then it is $\partial\bar{\partial}$ -exact. Equivalently, for every bidegree,*

$$\operatorname{im} \partial \cap \ker \bar{\partial} = \operatorname{im} \bar{\partial} \cap \ker \partial = \operatorname{im}(\partial\bar{\partial}).$$

More generally, if a form is d -exact and d -closed, then each of its (p, q) -components is a sum of a ∂ -exact form, a $\bar{\partial}$ -exact form, and a $\partial\bar{\partial}$ -exact form in the usual compatible way.

Remark 7.2 (What the lemma says in class). The statement is not that every closed form is $\partial\bar{\partial}$ -exact. It says that a form which is already exact in one complex direction and closed in the other direction has a primitive in both directions simultaneously. This is exactly the compatibility needed to make type decomposition independent of choices of representatives.

Proof idea in the Kähler case. The Kähler identities imply $\Delta_d = 2\Delta_{\bar{\partial}} = 2\Delta_{\partial}$ and hence identify all three harmonic theories. If $\alpha = \partial\beta$ and $\bar{\partial}\alpha = 0$, then the harmonic projection of α vanishes because it is ∂ -exact. The Green operator G therefore gives

$$\alpha = \Delta_{\bar{\partial}}G\alpha = \bar{\partial}\bar{\partial}^*G\alpha + \bar{\partial}^*\bar{\partial}G\alpha.$$

The condition $\bar{\partial}\alpha = 0$ removes the second term, and the Kähler commutation relations convert the remaining primitive into a $\partial\bar{\partial}$ primitive. The other formulations follow by conjugation and the same argument. $\square$

Example 7.3 (Why the non-Kähler distinction matters). On a general Hermitian manifold, Dolbeault Hodge theory still holds, but $\Delta_{\bar{\partial}}$ need not equal $\frac{1}{2}\Delta_d$. Thus a $\bar{\partial}$ -harmonic representative need not be d -harmonic, and the Green-operator proof above cannot be used. The conformally changed torus in Example 6.4 exhibits this failure explicitly.

7.2 The Dolbeault-to-de Rham spectral sequence

The decomposition by type gives a double complex

$$(\mathcal{A}^{p,q}(X), \partial, \bar{\partial}), \quad d = \partial + \bar{\partial}.$$

Filter the total complex by holomorphic degree, for example

$$F^p \mathcal{A}^k = \bigoplus_{r \geq p} \mathcal{A}^{r, k-r}.$$

The associated spectral sequence has

$$E_0^{p,q} = \mathcal{A}^{p,q}(X), \quad d_0 = \bar{\partial},$$

so that

$$E_1^{p,q} = H_{\bar{\partial}}^{p,q}(X), \quad d_1 = [\partial] : H_{\bar{\partial}}^{p,q}(X) \longrightarrow H_{\bar{\partial}}^{p+1,q}(X).$$

It converges to the associated graded pieces of de Rham cohomology:

$$E_{\infty}^{p,q} \cong \mathrm{Gr}_F^p H_{\mathrm{dR}}^{p+q}(X; \mathbb{C}).$$

This is the Frölicher spectral sequence.

Remark 7.4 (How to read the differentials). A class on the first page is represented by a $\bar{\partial}$ -closed form α . The first differential is represented by $\partial\alpha$. If $\partial\alpha = \bar{\partial}\beta$, then the class survives to the next page, and one continues by correcting the representative by lower-type terms. Thus higher differentials measure successive obstructions to finding a d -closed representative with prescribed leading (p, q) -term.

Theorem 7.5 (Kähler degeneration). *If X is compact Kähler, the Frölicher spectral sequence degenerates at E_1 . Consequently,*

$$H_{\mathrm{dR}}^k(X; \mathbb{C}) = \bigoplus_{p+q=k} H_{\bar{\partial}}^{p,q}(X), \quad h^{p,q} = h^{q,p}.$$

Classroom proof. Take a $\bar{\partial}$ -cohomology class and choose its $\bar{\partial}$ -harmonic representative α . The Kähler identities give $\Delta_d \alpha = 2\Delta_{\bar{\partial}} \alpha = 0$, hence $d\alpha = 0$ by the energy identity. Therefore $\partial\alpha = 0$ as well, so $d_1 = 0$. The same harmonic representative argument applies after every correction step, and all higher obstructions vanish. Hence $E_1 = E_{\infty}$. $\square$

Proposition 7.6 (The general inequality). *For every compact complex manifold,*

$$b_k \leq \sum_{p+q=k} h^{p,q}.$$

Equality for every k is equivalent to degeneration at E_1 .

7.3 A compact example: the Iwasawa manifold

The Iwasawa manifold is the standard compact example in which a holomorphic 1-form is not d -closed. Let G be the complex Heisenberg group

$$G = \left\{ \begin{pmatrix} 1 & z^1 & z^3 \\ 0 & 1 & z^2 \\ 0 & 0 & 1 \end{pmatrix} : z^1, z^2, z^3 \in \mathbb{C} \right\},$$

with multiplication induced by matrix multiplication. Let $\Gamma = G(\mathbb{Z}[i])$ and set

$$X = \Gamma \backslash G.$$

The quotient is a compact complex threefold: the lattice Γ acts by biholomorphisms, properly discontinuously, and cocompactly.

On $G \simeq \mathbb{C}^3$, consider the left-invariant holomorphic coframe

$$\varphi^1 = dz^1, \quad \varphi^2 = dz^2, \quad \varphi^3 = dz^3 - z^1 dz^2.$$

The transformation law under Γ preserves these forms, so they descend to global holomorphic 1-forms on X . Their structure equations are

$$d\varphi^1 = d\varphi^2 = 0, \quad d\varphi^3 = -dz^1 \wedge dz^2 = -\varphi^1 \wedge \varphi^2 \neq 0.$$

Since φ^3 is holomorphic, $\bar{\partial}\varphi^3 = 0$, while

$$\partial\varphi^3 = d\varphi^3 \neq 0.$$

Thus the Iwasawa manifold gives exactly the requested compact obstruction:

$$\boxed{\text{holomorphic 1-form} \not\Rightarrow d\text{-closed.}}$$

This also proves that X cannot admit a Kähler metric. Indeed, on a compact Kähler manifold every holomorphic 1-form is harmonic and hence d -closed, by the Kähler identity $\Delta_d = 2\Delta_{\bar{\partial}}$. The Iwasawa manifold therefore gives a direct differential-form obstruction, stronger in this context than merely checking Betti-number parity. It is also holomorphically parallelizable, so the failure is not caused by a lack of global holomorphic frames.

Remark 7.7 (Relation with the spectral sequence). For φ^3 , the Dolbeault class is not represented by a d -closed form with the same leading $(1,0)$ -term: its first Frölicher differential is represented by

$$d_1[\varphi^3] = [\partial\varphi^3] = -[\varphi^1 \wedge \varphi^2] \in H_{\bar{\partial}}^{2,0}(X).$$

Thus the non-closed holomorphic form is precisely a visible d_1 -obstruction in the Dolbeault-to-de Rham spectral sequence.

7.4 Deligne–Illusie: characteristic p and lifting

The Deligne–Illusie method proves degeneration by replacing analytic Hodge theory with an algebraic argument in positive characteristic.

Theorem 7.8 (Deligne–Illusie, simplified form). *Let X be a smooth proper scheme over a field of characteristic $p > 0$. Assume $\dim X < p$ and that X lifts to the ring of second Witt vectors $W_2(k)$. Then the Hodge-to-de Rham spectral sequence degenerates at E_1 . For a smooth proper complex variety, reduction modulo sufficiently large primes and the spreading-out/lifting argument imply degeneration in characteristic zero.*

The mechanism. The proof has four classroom steps:

1. The inverse Cartier transform identifies closed differential forms in characteristic p with Frobenius-twisted differential forms:

$$C^{-1} : \Omega_{X^{(p)}}^i \longrightarrow \mathcal{H}^i(F_*\Omega_X^\bullet).$$

2. A lift of Frobenius exists locally to $W_2(k)$. On overlaps, two lifts differ by a derivation; dividing the difference by p produces a homotopy term.
3. The local Cartier splittings glue up to homotopy. When the total degree is less than p , wedge products of the splittings define a quasi-isomorphism of truncated complexes:

$$\tau_{<p}(F_*\Omega_X^\bullet) \simeq \bigoplus_{i < p} \Omega_{X^{(p)}}^i[-i].$$

4. Taking hypercohomology gives a direct sum decomposition in degrees below p , so all differentials in the Hodge-to-de Rham spectral sequence vanish. Properness and dimension $< p$ then give full E_1 -degeneration.

Remark 7.9 (Why this is an approach, not a new Kähler identity). The method does not construct a Kähler metric and does not use analytic Laplacians. Its inputs are smoothness, properness, a W_2 -lifting, and the Cartier operator. The role of the dimension bound is precisely to keep all relevant degrees inside the range where the truncated Cartier splitting is available.

7.5 Applications of spectral-sequence degeneration

The vanishing of the higher differentials has several immediate consequences.

1. **Hodge decomposition.** For a compact Kähler manifold,

$$H_{\mathrm{dR}}^k(X; \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X).$$

2. **Betti-number inequality becomes equality.** Degeneration turns $b_k \leq \sum_{p+q=k} h^{p,q}$ into equality.
3. **Hodge symmetry.** Complex conjugation gives $h^{p,q} = h^{q,p}$, and Serre duality gives $h^{p,q} = h^{n-p, n-q}$ for a compact Kähler n -fold.
4. **Parity of odd Betti numbers.** The decomposition and conjugation pair the summands with $p \neq q$. The diagonal summand can occur only when $k = 2p$ is even; hence every odd Betti number of a compact Kähler manifold is even.
5. **Numerical obstructions.** An odd b_1 , as on a Hopf surface, rules out Kähler metrics. More generally, failure of the Frölicher equalities or of Hodge symmetry obstructs the existence of a Kähler structure.
6. **Algebraic degeneration.** Deligne–Illusie transfers the same E_1 -degeneration conclusion to smooth proper varieties in characteristic zero through reduction to characteristic p and the Cartier splitting.

Remark 7.10 (One-line synthesis).

$$\text{Kähler identities} \Rightarrow \partial\bar{\partial} \text{ lemma} \Rightarrow E_1\text{-degeneration} \Rightarrow \text{Hodge decomposition}.$$

The Deligne–Illusie route replaces the first analytic implication by

$$\text{Cartier operator} + W_2\text{-lifting} \Rightarrow E_1\text{-degeneration}.$$

8 Exercises

The first exercises fix coordinates and signs; the later ones test the theorem's hypotheses. All metrics use the normalization on the opening page. Solutions appear in the blackboard companion.

Exercise 1 (Types in two complex variables). On $\mathbb{C}^2$, decompose $dx_1 \wedge dx_2 + dy_1 \wedge dy_2$ into types. For $f = z_1^2 \bar{z}_2$, compute ∂f , $\bar{\partial} f$, $\partial \bar{\partial} f$, and $\bar{\partial} \partial f$. Explain why their last two values have opposite signs.

Hint. Use $dx = (dz + d\bar{z})/2$ and $dy = (dz - d\bar{z})/(2i)$; keep the order of wedge factors.

Exercise 2 (A torus Fourier mode). On $E = \mathbb{C}/(\mathbb{Z} + i\mathbb{Z})$ let $e_{m,n} = e^{2\pi i(mx+ny)}$. Compute $\partial e_{m,n}$ and $\bar{\partial} e_{m,n}$. Show that every smooth holomorphic function on E is constant by examining its Fourier coefficients. Explain why z itself is not a function on E although dz is a global form.

Hint. Use $\partial_z = (\partial_x - i\partial_y)/2$ and $\partial_{\bar{z}} = (\partial_x + i\partial_y)/2$.

Exercise 3 (Star and norm on an oblique elliptic curve). Let $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$, $v = \text{Im } \tau > 0$, with $g = |dz|^2$ and $z = x + \tau y$. Compute $*dz$, $*d\bar{z}$, $\int_{E_\tau} dz \wedge d\bar{z}$, and the L^2 Gram matrix of $dz, d\bar{z}$.

Hint. The complex-linear star on an oriented Euclidean plane satisfies $*dz = -i dz$.

Exercise 4 (A holomorphic change of frame). On the trivial holomorphic line bundle over $\mathbb{C}$, put $h(e, e) = e^{-|z|^2}$. Compute the Chern connection form and curvature in the frame e , and then in $e' = e e^z$. Verify the frame transformation formula directly.

Hint. For a line, $A = \partial \log h$ and $F = \bar{\partial} A$; $h' = |e^z|^2 h$.

Exercise 5 (Curvature on a topologically trivial torus line bundle). Give the trivial holomorphic line bundle on E the metric $h = e^{-u}$, where $u = \varepsilon \cos(2\pi x)$ and $\varepsilon \neq 0$. Compute A , F , and $\int_E iF/(2\pi)$. Explain how a trivial bundle can have a connection with nonzero curvature.

Hint. Compute $\partial \bar{\partial} u$ and use periodicity. A bundle and a connection are different pieces of data.

Exercise 6 (The hyperplane bundle on the sphere). For $\mathcal{O}(1)$ on $\mathbb{P}^1$, use $h = (1 + |z|^2)^{-1}$ on the affine chart. Compute F , prove $iF/(2\pi) = \omega_{\text{FS}}/\pi$, and integrate it. Check smooth compatibility with the chart $w = 1/z$ using a transition between holomorphic frames.

Hint. One may take $e_\infty = ze_0$ on the overlap. Integrate $r(1 + r^2)^{-2} dr d\theta$.

Exercise 7 (The tangent bundle of the sphere). For $g_{\text{FS}} = |dz|^2/(1 + |z|^2)^2$, compute the Chern connection and curvature of $T^{1,0}\mathbb{P}^1$ in the frame ∂_z . Show that its first Chern number is 2. How does the metric compare with the unit round sphere in Lecture 1?

Hint. The tangent Hermitian coefficient is $(1 + |z|^2)^{-2}$. Multiplying a metric by a positive constant does not change its Chern connection.

Exercise 8 (A hyperbolic quotient). Let $X = \Gamma \backslash \mathbb{H}$ be a compact hyperbolic surface of genus $g \geq 2$ with no orbifold points. From $g = y^{-2}|dz|^2$, compute the tangent connection form and show $c_1(TX) = -\omega_{\text{hyp}}/(2\pi)$. Use $\text{Area}(X) = 4\pi(g - 1)$ to recover the degree of TX and of its dual canonical bundle.

Hint. Differentiate $\log h = -2 \log y$, keeping $y_z = 1/(2i)$ and $y_{\bar{z}} = -1/(2i)$ distinct.

Exercise 9 (Conformal Kähler metrics). Let X have complex dimension n and let g_0 be Kähler. For $g_u = e^{2u}g_0$, compute $d\omega_u$. Show that every such metric is Kähler when $n = 1$, while for $n \geq 2$ it is Kähler only when u is locally constant. Give a periodic nonconstant example on a complex torus.

Hint. For $n \geq 2$, wedge $du \wedge \omega_0 = 0$ with ω_0^{n-2} and use that $\alpha \mapsto \alpha \wedge \omega_0^{n-1}$ is injective on one-forms.

Exercise 10 (Tangent torsion detects the Kähler condition). In holomorphic coordinates write $\omega = (i/2)G_{j\bar{k}}dz_j \wedge d\bar{z}_k$. Derive the coefficients of $\partial\omega$ and show that its vanishing is equivalent to symmetry of the Chern tangent coefficients in their two holomorphic indices. Explain why the conjugate condition then gives $d\omega = 0$.

Hint. Metric compatibility gives $\sum_a \Gamma_{ij}^a G_{a\bar{\ell}} = \partial_i G_{j\bar{\ell}}$.

Exercise 11 (Check a Kähler identity by hand). On $\mathbb{C}$ with $g = dx^2 + dy^2$, verify $[\Lambda, \partial](f d\bar{z}) = i\bar{\partial}^*(f d\bar{z})$. Compute $\Delta_d f$ and $\Delta_{\bar{\partial}} f$ and account for the factor of two.

Hint. Here $\Lambda(dz \wedge d\bar{z}) = -2i$ and $\bar{\partial}^*(f d\bar{z}) = -2f_z$.

Exercise 12 (Solving two torus equations). Let $f = \sum f_{m,n} e_{m,n}$ be smooth on the square elliptic curve. Give necessary and sufficient conditions for $\Delta_{\bar{\partial}} u = f$ and for $\bar{\partial} v = f d\bar{z}$ to have smooth solutions. Write the unique zero-mean solutions. Justify convergence of all derivatives, and compare the orders of the two denominators.

Hint. The nonzero eigenvalues of $\Delta_{\bar{\partial}}$ are $2\pi^2(m^2 + n^2)$. Smooth Fourier coefficients decay faster than every power.

Exercise 13 (The complex two-torus). For $X = \mathbb{C}^2/(\mathbb{Z}^2 + i\mathbb{Z}^2)$ with the flat metric, give bases of $\mathcal{H}^{1,0}$, $\mathcal{H}^{1,1}$, and $\mathcal{H}^{2,0}$. Determine all Hodge numbers and recover the real Betti numbers of the four-torus. Explain why nonzero Fourier modes contribute no harmonic forms.

Hint. The Laplacian acts coefficientwise in the constant frame.

Exercise 14 (Periods and changing complex structure). On the fixed smooth torus with period-one coordinates x, y , put $z = x + \tau y$. Express $[dx]$ and $[dy]$ in the basis $[dz], [d\bar{z}]$. Compute the periods of dz along the x - and y -cycles and identify the normalized top-degree class. Describe exactly what changes as τ varies in the upper half-plane.

Hint. Solve two linear equations and use $\int dz \wedge d\bar{z} = -2i \operatorname{Im} \tau$.

Exercise 15 (A wedge of harmonic forms need not be harmonic). On the square elliptic curve choose $g_u = e^{2u}(dx^2 + dy^2)$ with nonconstant periodic u . Show that dz and $d\bar{z}$ are harmonic but $dz \wedge d\bar{z}$ is not. Compute the harmonic representative of $[dx \wedge dy]$.

Hint. On a surface the star on one-forms is conformally invariant; harmonic top forms are constant multiples of the metric volume form.

Exercise 16 (A Hermitian metric where the Laplacians differ). On the flat complex two-torus set $g_u = e^{2u}g_0$ with $u = \varepsilon \cos(2\pi x_1)$ and $\varepsilon \neq 0$. Show that dz_1 is $\bar{\partial}$ -harmonic, but

$$d_{g_u}^* dz_1 = -2e^{-2u}u_{x_1} \neq 0.$$

Conclude that $\Delta_{d,g_u} \neq 2\Delta_{\bar{\partial},g_u}$. Explain why this does not imply that the complex torus admits no Kähler metric.

Hint. In real dimension 4, $*_{g_u} = e^{(4-2k)u}*_0$ on k -forms. For a $(1,0)$ -form, $\bar{\partial}^*$ has target type $(1,-1)$.

Exercise 17 (Duality with complex coefficients). On a compact Kähler n -fold, determine the types of $*\alpha$, $\bar{\alpha}$, and $*\bar{\alpha}$ for $\alpha \in \mathcal{H}^{p,q}$. Prove that integration gives a nondegenerate complex-bilinear pairing $H^{p,q} \times H^{n-p,n-q} \rightarrow \mathbb{C}$. Distinguish this pairing from the Hermitian L^2 inner product.

Hint. Choose the partner $*\bar{\alpha}$ and use positivity of $\int \alpha \wedge *\bar{\alpha}$.

Exercise 18 (When holomorphic forms are closed). Prove that every holomorphic p -form on a compact Kähler manifold is d -closed. Identify both the compactness and Kähler steps. Explain why $z_1 dz_2$ on $\mathbb{C}^2$ does not contradict the theorem.

Hint. A holomorphic $(p, 0)$ -form is $\bar{\partial}$ -harmonic; then use the Laplacian identity and the global energy identity.

Exercise 19 (The projective plane and the star). On $\mathbb{P}^2$ with $\omega = \omega_{\text{FS}}$, let $\eta = \omega/\pi$, so $\int_{\mathbb{P}^2} \eta^2 = 1$. Determine the Hodge diamond and the harmonic representatives of all cohomology classes. Show $*\omega = \omega$ and compute the intersection form on $H^2(\mathbb{P}^2, \mathbb{R})$.

Hint. In an oriented orthonormal real coframe, $\omega = e^1 \wedge e^2 + e^3 \wedge e^4$. Use the usual cohomology ring of projective space.

Exercise 20 (Two obstructions on a Hopf surface). Let $S = (\mathbb{C}^2 \setminus \{0\})/(z \sim 2z)$. Show that S is diffeomorphic to $S^3 \times S^1$ and that $|z|^{-2} \sum |dz_j|^2$ descends to a Hermitian metric. Give two proofs that S has no Kähler metric, using respectively b_1 and b_2 .

Hint. Use the coordinates $(z/|z|, \log |z| \bmod \log 2)$. If a Kähler form were exact, apply Stokes to its square.

Exercise 21 (Holomorphic differentials on a hyperbolic quotient). For $X = \Gamma \backslash \mathbb{H}$ as in Exercise 8, show that $f(z)dz$ descends precisely when $f(\gamma z)\gamma'(z) = f(z)$. If f is holomorphic, prove that the descended form is harmonic. Compute its L^2 norm over a fundamental domain and explain why the conformal factor y^{-2} cancels.

Hint. Pointwise $|dz|_g^2 = 2y^2$, while $dV_g = y^{-2} dx dy$.

Exercise 22 (Independence from the Kähler metric). Let g_0, g_1 be two Kähler metrics on a fixed compact complex manifold. Prove that the resulting subspaces $H^{p,q} \subset H_{\text{dR}}^{p+q}(X, \mathbb{C})$ agree by characterizing them through d -closed forms of pure type. Explain why the harmonic representatives themselves may still change. Use Exercise 15 for an explicit changing representative.

Hint. Apply the type-preserving harmonic projection for each metric to a d -closed pure-type representative.

References

- [1] Jean-Pierre Demailly, *Complex Analytic and Differential Geometry*. Author's freely available text: [author-hosted PDF](#). See the discussions of the Dolbeault lemma, Hermitian holomorphic bundles and the Chern connection, Kähler identities, and Hodge theory. Conventions for the fundamental form and the linearity of the Hermitian pairing should be compared with those fixed in this lecture before transferring formulas.
- [2] The Stacks Project, *Cohomology of a complex via an acyclic resolution*, [Tag 015E](#). This is the homological-algebra input behind the fine-resolution computation of Dolbeault cohomology, as in the de Rham comparison in Lecture 1.
- [3] Semyon Dyatlov, *18.155: Differential Analysis*, MIT, Fall 2021. [Course notes and lectures](#). The lectures on elliptic regularity, Fredholm operators, and elliptic operators on vector bundles provide the analytic background used here and in Lecture 1.
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